

Mind the Mediation Gap

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Abstract

Alignment is usually treated as optimization against a reward, preference, or rule-based objective. We study a different source of loss, which is the interface through which an aligned system is mediated. We introduce the Mediation Gap, separating unavoidable structural loss from finite-budget dynamic loss. Architectures induce local filtrations and feasible policy classes. Structural MG equals the minimum achievable oracle regret under those constraints. Dynamic MG becomes a quotient learning problem inside the fixed architecture. In a guardrail-mediated generation experiment, richer routing lowers structural MG but can increase realized finite-budget MG. The framework explains when alignment failures come from missing capacity versus unrealized capacity.

1 Introduction

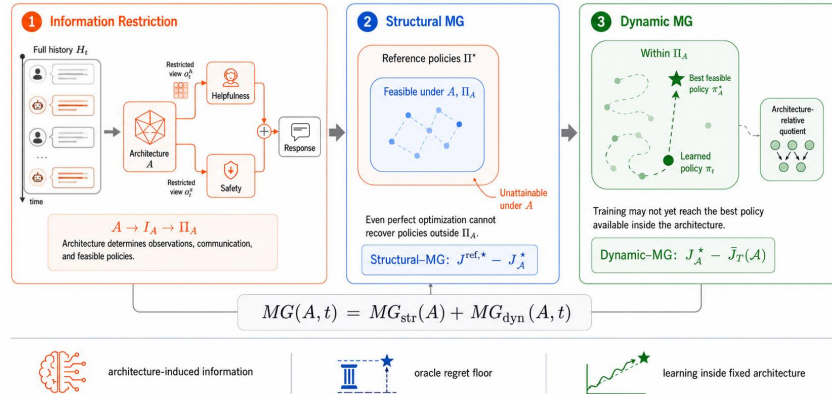


Figure 1: Overview of the Mediation Gap framework.

We study a basic question in reinforcement learning and alignment. How much performance is lost because an aligned system has to act through a mediated architecture. Current alignment methods are usually framed as optimization procedures. Given a reward function, a learned reward model, a preference objective, or a rule based surrogate, the algorithm searches for a policy with high expected value. RLHF [26], Constitutional AI [4], DPO [30], GRPO [33], and related pipelines largely follow this logic. The central difficulty is then taken to be estimation, optimization, robustness, and reward misspecification.

Modern systems introduce another difficulty. We do not directly control the internal state dynamics that implement aligned behavior. Instead, influence is mediated through interfaces. Reward models compress preferences into scalar feedback. Constitutional principles turn broad intent into checkable rules. Safety classifiers, verifiers, routers, and policy filters decide which signals are visible and which

actions are allowed. Alignment is therefore carried out through a hierarchy of mediated components. A high level goal must be interpreted, routed, and executed through a restricted architecture.

This paper formalizes the resulting loss as the *Mediation Gap* (MG). An architecture induces local information available to each component, hence a feasible policy class. The total gap separates into two terms. The *structural MG* is the value that no policy inside the mediated architecture can attain, even with perfect training. The *dynamic MG* is the value left unrealized because learning inside that architecture is finite and feedback limited. This split distinguishes impossibility from undertraining.

We develop the theory in three steps. First, we define architectures through filtrations and obtain policy class nesting from measurability. Second, we show that structural MG equals minimum achievable oracle regret under architectural constraints. Third, we study dynamic MG as a quotient learning problem. The learner need only identify the part of the environment that matters under the fixed architecture, while hidden value distinctions inside statistically indistinguishable cells can create persistent dynamic loss.

We believe the framework changes how we approach the long discussed alignment tax idea. A guardrail pipeline can fail because the safety evidence is absent, because it is present but poorly routed, or because training fails to exploit a sufficient interface. Our experiment instantiates this distinction in a sequential guardrail mediated generation task. We find that richer mediation can reduce structural MG while increasing finite budget dynamic MG. Thus a weaker architecture can look better after training because it is easier to saturate. The implication is operational. Alignment quality depends on both what the interface preserves and how learnable that preserved capacity is.

Related work. Our work is closest to work on information structure, partial observability, and alignment feedback. Altabaa and Yang study how information structure controls sample complexity in partially observable sequential teams and games [3]. Zhang et al. develop representation learning and planning guarantees for POMDPs [38]. These works analyze learnability under hidden state, while MG isolates how a delegated interface restricts what can be acted on and learned. In alignment, reward overoptimization shows that optimizing a learned feedback channel can diverge from true value [31], and recent methods add pessimism or stronger regularization to control this failure [18]. Reward model evaluation asks whether proxy scores predict downstream RLHF performance [10]. A more detailed discussion of related work and implications on modern systems can be found in Appendix B and C.

2 Preliminary

We begin from a simple observation. A capable system often acts through mediation. Full task context is available somewhere in the system, yet the final action is produced through modules, routers, critics, or specialized sub-policies that each see only part of that context. Once this happens, performance can fall for two very different reasons. One reason comes from the architecture itself. The architecture determines what each agent can observe, what messages can be exchanged, and what information can be retained over time. The second reason comes from learning. Even within a fixed architecture, training may fail to reach the best policy available in that architecture.

Running Example. Consider a guardrail-mediated generation pipeline. A user query and its conversational context induce a full history H_t , as shown in Figure 1. A mediator routes restricted views to two delegates. The *Helpfulness* model represents the harmless assistant that tries to answer the user query. The *Safety* model represents a safeguard model (e.g., constitutional classifier, policy filter). The mediator is the orchestration layer that decides what each component sees, whether clarification is available, and how safety evidence affects the final response. If this interface preserves and routes the decision-relevant evidence, the mediated system can match the reference response. If it loses or blocks that evidence, a structural mediation gap remains even under perfect training. If the interface is sufficient but training fails to use it, the gap is dynamic. This running example will reappear throughout the paper. It gives an intuitive interpretation of the theory and the experiment.

2.1 Architectures and Policy Classes

We formulate the framework on histories. Let H_t denote the full history up to time t . An architecture $\mathcal{A}$ induces, for each agent m and time t , an information set represented by a filtration $\mathcal{I}_t^{m,\mathcal{A}} \subseteq \sigma(H_t)$. This induces a policy class

$$\Pi_{\mathcal{A}} = \{\pi : \pi_t^m \text{ is } \mathcal{I}_t^{m,\mathcal{A}}\text{-measurable for all } m, t\}.$$

The task objective is described by a single performance functional J . This objective remains fixed throughout the framework. We also fix a reference policy class Π^{ref} , which provides the benchmark $J^{\text{ref},*} = \sup_{\pi \in \Pi^{\text{ref}}} J(\pi)$. Then for any architecture $\mathcal{A}$, the best value achievable inside its induced class is $J_{\mathcal{A}}^* = \sup_{\pi \in \Pi_{\mathcal{A}}} J(\pi)$.

2.2 Decomposition

Suppose a learning procedure under architecture $\mathcal{A}$ produces policies $\pi_1, \dots, \pi_T \in \Pi_{\mathcal{A}}$. We define the average true performance by

$$\bar{J}_T(\mathcal{A}) = \frac{1}{T} \sum_{t=1}^T J(\pi_t),$$

which is the average value of the realized policies under the target objective J . The Mediation Gap relative to the reference class is

$$\text{MG}_T(\mathcal{A} \mid \Pi^{\text{ref}}) = J^{\text{ref},*} - \bar{J}_T(\mathcal{A}) = \underbrace{J^{\text{ref},*} - J_{\mathcal{A}}^*}_{\text{structural}} + \underbrace{J_{\mathcal{A}}^* - \bar{J}_T(\mathcal{A})}_{\text{dynamic}}.$$

The first term depends only on the architecture through its induced policy class. It measures what the architecture makes unattainable even under ideal optimization. The second term depends on the realized learning process inside that class. It measures the gap between what the delegated architecture can express and what training actually realizes. This identity is the backbone of the MG framework. In the toy model (Figure 1), if the Helpfulness and Safety views lose information needed for the best response, the loss is structural. If that information is preserved but training still fails to use it well, the loss is dynamic.

2.3 Nesting

Architectures can be ordered by the information they provide. We write $\mathcal{A}_2 \preceq \mathcal{A}_1$ when every agent under $\mathcal{A}_1$ has access to an information filtration that contains the one induced by $\mathcal{A}_2$, that is,

$$\mathcal{I}_t^{m, \mathcal{A}_2} \subseteq \mathcal{I}_t^{m, \mathcal{A}_1} \quad \text{for all } m, t.$$

This order matters because information inclusion induces policy class inclusion. The formal statement is given in Proposition A.1 in the appendix. The implication is intuitive. If a richer architecture gives every agent at least the information available under a poorer one, then every policy admissible under the poorer architecture can be implemented under the richer architecture as well.

This principle provides a clean basis for structural ladders. If $\mathcal{A}^{(0)} \preceq \mathcal{A}^{(1)} \preceq \dots \preceq \mathcal{A}^{(L)}$, then $\Pi_{\mathcal{A}^{(0)}} \subseteq \Pi_{\mathcal{A}^{(1)}} \subseteq \dots \subseteq \Pi_{\mathcal{A}^{(L)}}$, which implies $J_{\mathcal{A}^{(0)}}^* \leq J_{\mathcal{A}^{(1)}}^* \leq \dots \leq J_{\mathcal{A}^{(L)}}^*$.

Along such a ladder, the structural gap telescopes exactly:

$$J_{\mathcal{A}^{(L)}}^* - J_{\mathcal{A}^{(k)}}^* = \sum_{\ell=k+1}^L \left(J_{\mathcal{A}^{(\ell)}}^* - J_{\mathcal{A}^{(\ell-1)}}^* \right).$$

Two common sources of architectural order are important in the sequel. One comes from observation design. If one observation process is obtained as a measurable coarsening of another, the induced policy classes are nested. The other comes from coordination design. If one communication graph extends another and the richer protocol can emulate the poorer one, the induced policy classes are again nested. See Propositions A.2 and A.3 for formal statement.

3 Structural Capacity

We now quantify structural value loss from architectural information constraints. Classical RL work studies compression within a fixed architecture. State abstraction characterizes when states can be merged without value loss [1, 2]; value equivalence identifies when compressed models preserve Bellman structure [13, 14]. Here, the architecture itself is the constraint, which determines which history distinctions are observable before any compression or learning begins.

3.1 Running Example

A full prompt history H is mapped into two mediated views,

$$Z^{\text{help}} = \phi^{\text{help}}(H), \quad Z^{\text{safe}} = \phi^{\text{safe}}(H).$$

115 A *Helpfulness* delegate reads Z^{help} . A *Safety* delegate reads Z^{safe} . Their mediated interaction
 116 produces the final action A . Let $U(a, h)$ be the utility of action a at history h , and let $V^*(h) :=$
 117 $\max_a U(a, h)$ be the full information optimum in this one step problem.

118 To separate information loss from coordination details, let $\mathcal{Q}_\phi(z) \subseteq \Delta(\mathcal{A})$ denote the set of action
 119 distributions realizable after the mediated signal $Z = z$ has been produced. The best delegated value
 120 is

$$J_\phi^* = \mathbb{E} \left[\sup_{q \in \mathcal{Q}_\phi(Z)} \sum_a q(a) \mathbb{E}[U(a, H) \mid Z] \right].$$

121 **Proposition 1** (Conditional delegated regret). *Define*

$$\delta_\phi(z) = \mathbb{E}[V^*(H) \mid Z = z] - \sup_{q \in \mathcal{Q}_\phi(z)} \sum_a q(a) \mathbb{E}[U(a, H) \mid Z = z].$$

122 *Then $\mathbb{E}[V^*(H)] - J_\phi^* = \mathbb{E}[\delta_\phi(Z)]$.*

123 **Remark.** Structural loss is exactly the Bayes regret induced by the mediated information cell. If
 124 the mediated views erase distinctions that matter for the reference decision, then the regret remains
 125 positive even under perfect optimization. When $\mathcal{Q}_\phi(z) = \Delta(\mathcal{A})$ for every z , the second term reduces
 126 to the ordinary Bayes action value $\max_a \mathbb{E}[U(a, H) \mid Z = z]$.

127 3.2 Regret Identity

128 We now extend the model to the general adaptive setting. Let H_t be the full history at time t . Let
 129 A_t be the action selected by a delegated policy $\pi \in \Pi_a$. The reference benchmark from Section 2 is
 130 the centralized causal class with full history access. Its Bellman optimal value functions are written
 131 $V^*(h_t), Q^*(h_t, a)$. Define the one step oracle regret $d^*(h_t, a) = V^*(h_t) - Q^*(h_t, a)$.

132 **Theorem 1** (Oracle regret identity). *For every delegated policy $\pi \in \Pi_a$,*

$$J^* - J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

133 Theorem 1 works directly on adaptive trajectories. It places the architectural question on the same
 134 footing as the infinite horizon control problem itself. A delegated policy incurs structural loss exactly
 135 through the discounted oracle regret accumulated along the histories that it generates.

136 **Corollary 1** (Structural regret). *For every architecture a ,*

$$J^* - J_a^* = \inf_{\pi \in \Pi_a} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

137 Corollary 1 connects to the structural term from Section 2. The architecture enters only through the
 138 information it makes available. Sparse communication, restricted observations, and limited memory
 139 all become instances of the same object. They change the feasible class Π_a , and the resulting loss is
 140 the smallest oracle regret compatible with that class.

141 3.3 Capacity Curve

142 Let $\mathcal{A}(R)$ denote the family of causal architectures feasible under budget R . The budget may encode
 143 communication rate, message alphabet size, memory, observation complexity, or a joint resource
 144 constraint over several such dimensions. Larger budgets enlarge the feasible family. Define

$$\text{MG}_{\text{str}}(R) = J^* - \sup_{a \in \mathcal{A}(R)} J_a^*.$$

145 **Theorem 2** (Endogenous mediation capacity). *The structural capacity curve satisfies*

$$\text{MG}_{\text{str}}(R) = \inf_{a \in \mathcal{A}(R)} \inf_{\pi \in \Pi_a} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

Remark. Theorem 2 gives the best oracle regret achievable under a constrained causal architecture. The term endogenous refers to the feedback loop created by the architecture. The architecture determines the policy class, the policy class shapes the trajectories, and regret is measured on those trajectories.

Corollary 2 (Zero loss criterion). $\text{MG}_{\text{str}}(R) = 0$ iff

$$\inf_{a \in \mathcal{A}(R)} \inf_{\pi \in \Pi_a} \mathbb{E}_{\pi} \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right] = 0.$$

When the infimum is attained, this holds if and only if $d^*(H_t, A_t) = 0$ almost surely for every t under some $a \in \mathcal{A}(R)$ and $\pi \in \Pi_a$.

The one step running example from Section 3.1 is the static specialization of Theorem 2. There the trajectory has one step, the discounted sum has one term, and structural MG reduces to conditional Bayes regret under mediated views.

4 Dynamic Learning Capacity

We now turn to the second half of the MG. Once an architecture is fixed, the structural term from Section 2 and Section 3 is fixed as well. What remains is a learning problem inside the delegated interface. This distinction is central in modern alignment. A reward model, a judge, a verifier, or a safety filter does more than score outputs. It determines which task distinctions can ever be revealed through feedback. A converse asks what uncertainty survives this interface. An upper bound asks for an explicit learner that extracts the right distinctions through the same interface while keeping the information budget under control. Recent work on information structure in partially observable reinforcement learning, representation learning for POMDPs, and reward overoptimization in alignment makes this theme concrete [11, 31].

As we discussed, structural MG asks what the architecture erases before deployment begins. Dynamic MG asks what the learner still has to discover after the architecture has been fixed.

4.1 Running Example

A hidden environment variable Θ captures a latent mode of the task. One may read Θ as a hidden user type, a hidden safety regime, or a hidden normative distinction in the prompt distribution. The architecture a fixes a delegated policy class Π_a . In each round we deploy one policy $\pi \in \Pi_a$, receive an observable feedback signal Y , and evaluate the resulting episode by its true value $J_{\Theta}(\pi)$.

Suppose $\Theta \in \{\theta^+, \theta^-\}$ under prior μ_0 . Let K_{θ}^{π} denote the law of the observable feedback Y when policy π is deployed in environment θ . Let $J_{a,\theta}^* = \sup_{\pi \in \Pi_a} J_{\theta}(\pi)$. Assume the architecture induces the same observable feedback law in both latent modes, $K_{\theta^+}^{\pi} = K_{\theta^-}^{\pi}$ for every $\pi \in \Pi_a$.

Define the Bayes gap of the repeated toy by

$$\delta_a^{\text{toy}} = \inf_{\pi \in \Pi_a} \sum_{s \in \{+, -\}} \mu_0(\theta^s) (J_{a,\theta^s}^* - J_{\theta^s}(\pi)).$$

Proposition 2 (Dynamic floor). If $K_{\theta^+}^{\pi} = K_{\theta^-}^{\pi}$ for every $\pi \in \Pi_a$, then for every learner $\mathbb{L}$ and every horizon T , $\mathcal{D}_{a,T}(\mathbb{L}; \mu_0) \geq \delta_a^{\text{toy}}$. If no single policy in Π_a attains $J_{a,\theta}^*$ for both $\theta = \theta^+$ and $\theta = \theta^-$, then $\delta_a^{\text{toy}} > 0$.

Remark. Proposition 2 shows that even after repeated interaction, the learner never receives evidence that separates θ^+ from θ^- . If those two latent modes induce different best delegated policies, a positive regret floor remains forever. This is already an alignment problem. A reward model or a judge can produce the same visible signal on two cases whose best aligned responses differ. Then training keeps moving inside an observation cell that is statistically stable while still carrying unresolved value ambiguity.

4.2 Bayesian Learning

We now pass to the general model. Fix an architecture a , hence a feasible policy class Π_a . Each round n is one deployment episode. Inside that episode the deployed policy runs for the full discounted

control problem. Its value in environment θ is

$$J_\theta(\pi) = \mathbb{E}_{\theta, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r_\theta(H_t, A_t) \right].$$

Now the connection is clear. Section 3 studies value loss inside an episode. Section 4 studies how repeated episodes reveal the environment well enough to recover the best policy available inside the same architecture.

Let $\Theta \sim \mu_0$ be an unknown environment. At round n , the learner selects $\pi_n \in \Pi_a$ using the past transcript $F_{n-1} = (\pi_1, Y_1, \dots, \pi_{n-1}, Y_{n-1})$, then receives $Y_n \sim K_{\Theta}^{\pi_n}$. The class optimal value is $J_{a, \theta}^* = \sup_{\pi \in \Pi_a} J_\theta(\pi)$, and the dynamic MG over T rounds is

$$\mathcal{D}_{a, T}(\mathbb{L}; \mu_0) = \frac{1}{T} \mathbb{E}_{\Theta \sim \mu_0, \mathbb{L}} \left[\sum_{n=1}^T (J_{a, \Theta}^* - J_\Theta(\pi_n)) \right].$$

This definition isolates the learning problem that remains after structural limits have already been fixed. We compare the learner with an oracle that knows the environment while still respecting the delegated class Π_a . Dynamic MG is therefore the price of uncertainty inside a restricted architecture. In reinforcement learning terms, it is a Bayes regret problem under partial observability. Intuitively, it is the price of having to infer the task relevant distinctions that the feedback pipeline actually exposes.

4.3 Two Quotients

The environment class is usually much larger than the learning problem. A learner inside architecture a has no use for distinctions between environments that never change any feasible deployment outcome. This leads to two quotients.

The first quotient captures what interaction can reveal. Define the statistical relation

$$\theta \approx_a \theta' \iff K_{\theta}^{\pi} = K_{\theta'}^{\pi} \quad \text{for every } \pi \in \Pi_a.$$

Let $Z_a^{\text{stat}} = \Theta / \approx_a$. Two environments lie in the same statistical cell when every feasible deployment produces the same feedback law under both.

The second quotient captures what regret depends on. Define the joint relation

$$\theta \sim_a \theta' \iff \theta \approx_a \theta' \text{ and } \forall \pi \in \Pi_a, J_\theta(\pi) = J_{\theta'}(\pi).$$

Let $Z_a = \Theta / \sim_a$. Thus Z_a refines Z_a^{stat} . The first quotient is about observable evidence. The second is about observable evidence together with decision value.

Theorem 3 (Statistical factorization). *For every learner $\mathbb{L}$ and every horizon T , the law of the transcript F_T depends on Θ only through Z_a^{stat} . Equivalently,*

$$P(F_T \in \cdot \mid \Theta = \theta) = P(F_T \in \cdot \mid Z_a^{\text{stat}} = [\theta]_{\text{stat}}).$$

Hence $I(\Theta; F_T) = I(Z_a^{\text{stat}}; F_T) \geq I(Z_a; F_T)$.

So interaction can only target the statistically visible quotient. This is the object that the feedback channel can reveal. Even so, the learner may still care about a finer partition of environments if value differs inside one statistical cell.

Theorem 4 (Joint quotient reduction). *Let T_a be the quotient map from Θ to Z_a , and let $T_{a\#}\mu_0$ be the pushforward prior on Z_a . Then for every learner $\mathbb{L}$,*

$$\mathcal{D}_{a, T}(\mathbb{L}; \mu_0, \Theta) = \mathcal{D}_{a, T}(\mathbb{L}; T_{a\#}\mu_0, Z_a).$$

Remark. Theorem 4 is the dynamic analogue of the reduction principle from Section 3. There, the architecture compresses the policy space down to what is feasible. Here, the same architecture compresses the environment space down to what is decision relevant for that feasible class.

4.4 Hidden Value Uncertainty

Once the quotient is defined, the natural state variable for learning is the posterior over joint quotient cells. Let $\nu_n = \mathbb{P}(Z_a \in \cdot \mid F_n)$. For a posterior ν on Z_a , define the Bayes envelope

$$\delta_a(\nu) = \inf_{\pi \in \Pi_a} \int (J_a^*(z) - J_z(\pi)) \nu(dz),$$

where $J_z(\pi)$ is well defined because all environments in the same joint cell agree on the values of feasible policies.

227 **Theorem 5** (Bayes envelope). *For every learner $\mathbb{L}$,*

$$\mathcal{D}_{a,T}(\mathbb{L}; \mu_0) \geq \frac{1}{T} \sum_{n=1}^T \mathbb{E}[\delta_a(\nu_{n-1})].$$

228 The envelope has the same role here that oracle regret had in Section 3. It is the smallest one round
 229 Bayes regret still forced by the learner’s current uncertainty. When the posterior spreads mass over
 230 joint cells that prescribe different best policies, dynamic MG stays positive.

231 The two quotient split reveals a second phenomenon. Feedback may separate statistical cells while
 232 still leaving value ambiguity inside each such cell. Let g_a be the refinement map from Z_a to Z_a^{stat} .
 233 For a statistical cell $z \in Z_a^{\text{stat}}$, let $\Xi_0(\cdot | z)$ be the prior conditional law of the joint quotient inside
 234 the fiber $g_a^{-1}(z)$. Define

$$b_a(z) = \inf_{\pi \in \Pi_a} \int_{g_a^{-1}(z)} (J_a^*(z') - J_{z'}(\pi)) \Xi_0(dz' | z).$$

235 **Theorem 6** (Hidden value floor). *For every learner $\mathbb{L}$ and every horizon T ,*

$$\mathcal{D}_{a,T}(\mathbb{L}; \mu_0) \geq \mathbb{E}[b_a(Z_a^{\text{stat}})].$$

236 Theorem 6 says that interaction can fully exhaust the statistically visible part of the problem while a
 237 positive regret floor still remains. The remaining gap comes from value distinctions that the feedback
 238 channel never reveals. In alignment language, the learner may infer everything that the judge, reward
 239 model, or safety signal is capable of expressing, while the true objective still varies inside that visible
 240 cell. Appendix F.8 records a simple collapse criterion. If episode value is itself readable from
 241 feedback, then the two quotients coincide and this floor disappears.

242 4.5 Endogenous Learning Capacity

243 Section 3 closed with a capacity curve over architecture budgets. The dynamic analogue is a capacity
 244 curve over information acquired through interaction. Since the transcript factors through Z_a^{stat} , this
 245 is the natural variable to budget. We define the converse frontier

$$\underline{\text{ELC}}_a(T, B) = \inf_{\mathbb{L}, I(Z_a^{\text{stat}}; F_T) \leq B} \frac{1}{T} \sum_{n=1}^T \mathbb{E}[\delta_a(\nu_{n-1})],$$

246 and the attainable frontier

$$\text{ELC}_a^{\text{ach}}(T, B) = \inf_{\mathbb{L}, I(Z_a^{\text{stat}}; F_T) \leq B} \mathcal{D}_{a,T}(\mathbb{L}; \mu_0).$$

247 These two objects play different roles. The converse frontier is dictated by posterior geometry. The
 248 attainable frontier is an algorithmic object. It asks for an explicit learner that extracts the right
 249 information through the delegated feedback channel while keeping the total information acquisition
 250 below B . By Theorem 5, $\text{ELC}_a^{\text{ach}}(T, B) \geq \underline{\text{ELC}}_a(T, B)$. Appendix F.9 gives one generic route
 251 toward an upper bound. If a learner can maintain a favorable regret to information tradeoff at
 252 each round, then the attainable frontier scales as $\sqrt{B/T}$. This is where design matters, as a lower
 253 bound identifies what the learner must know, and an upper bound asks how to uncover exactly that
 254 information through the architecture that mediation has already imposed.

255 5 Mediation Gap Under a Surrogate Model

256 We now give a closed form slice of the framework, it shows how the structural and dynamic terms
 257 become explicit when the mediated decision problem is approximated by a quadratic surrogate. Write
 258 cost as $L = -J$. Let X denote the full local episode state and let $Z = \phi(X)$ denote the mediated
 259 view. Around the operating regime of interest, take

$$\tilde{\ell}_X(u) = \frac{1}{2} u^\top H u + h(X)^\top u, H = H^\top \succ 0.$$

260 Sparse coordination is encoded by a subspace $\text{Im}(P_E)$. Define

$$K_E = P_E(P_E^\top H P_E)^{-1} P_E^\top, m(Z) = \mathbb{E}[h(X) | Z].$$

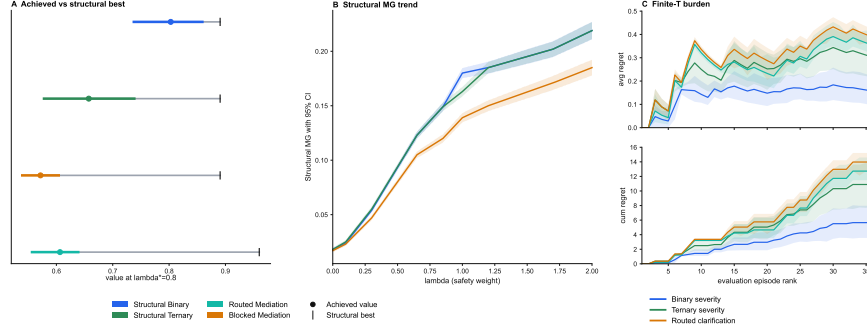


Figure 2: Finite budget behavior. Panel A compares achieved value with structural best at $\lambda = 0.8$. Panel B shows how structural MG changes with the safety weight. Panel C traces the finite budget burden across evaluation episodes.

261 The dense, sparse, and partial surrogate optima are

$$u_d(X) = -H^{-1}h(X), u_s(X) = -K_E h(X), u_p(Z) = -K_E m(Z).$$

262 The surrogate structural MG splits exactly into a coordination term and an information term:

$$C = \frac{1}{2} \mathbb{E}[h(X)^\top (H^{-1} - K_E) h(X)], \quad I = \frac{1}{2} \mathbb{E}[(h(X) - m(Z))^\top K_E (h(X) - m(Z))].$$

263 Thus coordination loss is the price of forcing the control into the sparse subspace. Information loss is
264 the decision weighted conditional variance left after mediation.

265 The same surrogate also gives a dynamic target. If a learner outputs $\hat{u}_T(Z) = -K_E \hat{m}_T(Z)$, then

$$D_T^{\text{quad}} = \tilde{L}(\hat{u}_T) - \tilde{L}(u_p) = \frac{1}{2} \mathbb{E}[(\hat{m}_T(Z) - m(Z))^\top K_E (\hat{m}_T(Z) - m(Z))].$$

266 So dynamic MG is estimation error for the quotient statistic $m(Z)$, or more precisely $P_E^\top m(Z)$.

267 Let u_J be the true dense optimum and let $\delta(u) = L(u) - \tilde{L}(u)$. Then

$$L(\hat{u}_T) - L(u_J) = A + C + I + D_T^{\text{quad}} + \Xi_{A \otimes C} + \Xi_{A \otimes I} + \Xi_{A \otimes D}.$$

268 Under the local accuracy condition $|\delta(u)| \leq \eta$ on the relevant controllers, $L(\hat{u}_T) - L(u_J) \leq$
269 $C + I + D_T^{\text{quad}} + 8\eta$. The full version of this section is given in Appendix G.

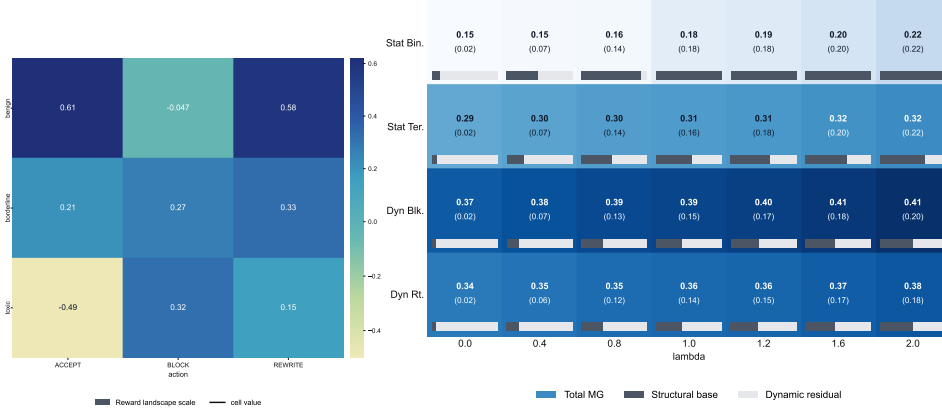
270 6 Experiment

271 **Setup.** We test the framework in a guardrail mediated generation task using RealToxicityPrompts
272 [12] with Qwen3.6 and Qwen3Guard [28, 39]. Each episode contains a user prompt, a hid-
273 den task mode, an initial safety signal, an optional clarification step, and a final action in
274 {ACCEPT, REWRITE, BLOCK}. (Code)

275 The scalar reward, action set, prompt pool, and evaluation stack are fixed. Only the mediation
276 interface changes. We compare four architectures. Static Binary sees a coarse safety signal. Static
277 Ternary sees a finer severity signal. Dynamic Blocked may request clarification, though the answer
278 cannot affect the final action. Dynamic Routed may request clarification and route the answer to the
279 final decision. An oracle has direct access to the latent mode.

280 **Training logic.** For each architecture we enumerate its feasible policy tables. We then precompute
281 a counterfactual outcome tensor over episodes, final actions, and clarification visibility. This lets
282 every architecture be evaluated against the same response, guard, and judge outcomes. The structural
283 mediation gap is estimated by exhaustive search over the architecture feasible policy class. The
284 dynamic mediation gap is estimated by training a finite budget learner inside that class and comparing
285 the achieved value with the architecture’s own structural best. Thus the experiment separates what
286 the interface permits from what training actually realizes.

287 **Decision landscape.** Figure 3a shows why mediation matters. Accept is valuable for benign prompts
288 and costly for toxic prompts. Rewrite is strongest in the ambiguous region. Extra information is useful
289 only when it preserves distinctions that change the best action. This is why the cost of coarsening
290 rises as safety receives more weight.



(a) Reward landscape.

(b) MG decomposition map.

Figure 3: Results summary.

Capacity versus realization. Figure 2 shows the central empirical pattern. Structural MG grows with the safety weight, which confirms that missing safety relevant distinctions becomes more costly when unsafe mistakes are penalized more strongly. Yet richer mediation does not automatically improve finite budget performance. Static Binary is weak but easy to saturate. Dynamic Routed and Dynamic Blocked preserve more capacity, yet the learner leaves much of that capacity unused. This is the empirical reason for separating structural MG from dynamic MG.

Decomposition map. Figure 3b gives the compressed view. Each cell reports total realized MG against the oracle. The number in parentheses is the structural base. The remaining mass is the dynamic residual. Routed clarification has a smaller structural base than blocked clarification, which shows that evidence must reach the decision point. Yet routed clarification still has a large residual, which shows that preserving useful information is different from learning how to use it.

Insights. The experiment gives four lessons. First, weaker mediation can look better after training because it is easier to saturate. Second, architectural richness and finite budget performance can move in opposite directions. Third, routing is distinct from observation. Clarification has little value if its answer cannot affect the final action. Fourth, information matters only when it protects action changing distinctions. In this setting, the alignment tax is therefore partly structural and partly learnability driven. The structural part comes from the mediated interface. The dynamic part comes from unrealized capacity inside that interface.

7 Conclusion

We introduced the Mediation Gap as a framework for understanding alignment loss created by mediated architectures. Structural MG measures value lost because an interface removes or misroutes decision relevant information. Dynamic MG measures value left unrealized because learning inside the mediated class is finite. This distinction clarifies the alignment tax in modern systems. Some tax comes from the interface itself, such as reward models, verifiers, safety classifiers, routers, and policy filters. Some comes from the difficulty of learning how to use that interface. Our theory formalizes architectures through information filtrations, gives structural MG an oracle regret interpretation, and treats dynamic MG as quotient learning inside a fixed architecture.

The framework not only provides theoretical insights for LLM safety research, but also applies whenever a principal objective must be pursued through restricted observations, delegated modules, or limited coordination. Our experiment shows the practical relevance of the split. A richer interface can preserve more value in principle yet perform worse after training because its capacity is harder to realize. Alignment design must therefore optimize both mediation structure and learnability.

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A Notation Guide

We begin by refreshing the main notation used throughout the paper. The structural capacity curve and the finite budget mediation gap are two levels of the same framework. The first is an ex-ante architecture design frontier. The second is the realized gap of one trained mediated system.

Reference benchmark. Let Π^{ref} be the reference policy class and let

$$J^{\text{ref},*} = \sup_{\pi \in \Pi^{\text{ref}}} J(\pi)$$

be the value of the reference system. The reference class represents the benchmark available before mediation constraints are imposed. In the experiment, the oracle architecture plays this role inside the finite policy abstraction.

Architecture induced policy class. An architecture A induces local information for each component and therefore a feasible policy class Π_A . Its best attainable value is

$$J_A^* = \sup_{\pi \in \Pi_A} J(\pi).$$

This value assumes perfect optimization inside the mediated class. It measures what the architecture permits in principle.

Finite budget realized value. Let $\bar{J}_T(A)$ denote the value achieved by a learner after T training rounds inside architecture A . Since every deployed policy lies in Π_A , we have

$$\bar{J}_T(A) \leq J_A^*.$$

The gap $J_A^* - \bar{J}_T(A)$ is the unrealized capacity left by finite training.

Structural capacity frontier. For a resource budget R , let $\mathcal{A}(R)$ be the set of feasible architectures under that budget. The structural mediation gap frontier is

$$\text{MG}_{\text{str}}(R) = J^{\text{ref},*} - \sup_{a \in \mathcal{A}(R)} J_a^*.$$

This is an architecture design object. It asks how small the structural gap can be when one may choose the best architecture under budget R . It contains no learning term.

Finite budget mediation gap. For a fixed architecture A , the realized finite budget mediation gap is

$$\text{MG}_T(A \mid \Pi^{\text{ref}}) = J^{\text{ref},*} - \bar{J}_T(A).$$

It decomposes exactly as

$$\text{MG}_T(A \mid \Pi^{\text{ref}}) = \underbrace{J^{\text{ref},*} - J_A^*}_{\text{structural MG}} + \underbrace{J_A^* - \bar{J}_T(A)}_{\text{dynamic MG}}.$$

The structural term is the loss caused by the mediated interface. The dynamic term is the loss caused by finite training inside that interface.

Relation between the two quantities. If $A \in \mathcal{A}(R)$, then

$$\text{MG}_T(A \mid \Pi^{\text{ref}}) \geq J^{\text{ref},*} - J_A^* \geq \text{MG}_{\text{str}}(R).$$

The first inequality follows from $\bar{J}_T(A) \leq J_A^*$. The second follows because $\sup_{a \in \mathcal{A}(R)} J_a^* \geq J_A^*$. Hence $\text{MG}_{\text{str}}(R)$ is the best structural lower envelope available under budget R , while $\text{MG}_T(A \mid \Pi^{\text{ref}})$ is what a particular trained system actually pays.

When the quantities coincide. The two quantities coincide in the idealized limit where the selected architecture is structurally optimal under the budget and learning reaches the best value inside that architecture. Formally, if

$$A \in \arg \max_{a \in \mathcal{A}(R)} J_a^* \quad \text{and} \quad \bar{J}_T(A) \rightarrow J_A^*,$$

then

$$\text{MG}_T(A \mid \Pi^{\text{ref}}) \rightarrow \text{MG}_{\text{str}}(R).$$

Away from this limit, the realized finite budget gap can be much larger than the structural frontier.

Notation caution. We use A for a single architecture and $\mathcal{A}(R)$ for a family of feasible architectures under resource budget R . If a family $\mathcal{A}$ is optimized jointly, define

$$J_{\mathcal{A}}^* = \sup_{a \in \mathcal{A}} J_a^*.$$

If training also chooses the architecture, then $\bar{J}_T(\mathcal{A})$ must specify whether the learner selects policies only, architectures only, or both. In this paper, $\text{MG}_T(A \mid \Pi^{\text{ref}})$ refers to a fixed architecture unless stated otherwise.

Interpretation for the experiment. The experiment estimates J_a^* by exhaustive search over the finite policy class induced by each mediated architecture. It estimates $\bar{J}_T(a)$ by training a finite budget learner inside that same class. The reported structural MG is the plug in estimate of $J^{\text{ref},*} - J_a^*$. The reported dynamic MG is the residual $J_a^* - \bar{J}_T(a)$. This mirrors the theoretical decomposition while keeping the reference class, architecture class, and learner fixed within each comparison.

B More Related Work

Information structures and team decision theory. MG builds on the classical view that information patterns determine what a distributed decision system can implement. Blackwell compares experiments by their decision value, not by raw entropy [5]. Marschak and Radner, Radner, and Ho and Chu study team decisions under decentralized information [16, 23, 29]. Witsenhausen’s counterexample shows that decentralized information can create irreducible difficulty even in simple quadratic settings [37]. Common information methods later gave a way to reduce some decentralized problems to centralized belief states [24]. Our contribution is complementary. We use filtrations to define the mediated policy class directly, then express loss as a structural MG and a dynamic MG relative to that class.

State abstraction and decision-relevant representation. Approximate state abstraction and value equivalence study which state or model distinctions can be removed while preserving decision quality [1, 2, 13, 14]. MG applies the same decision-relevance principle to mediated architectures. The relevant object is not only a compressed state. It is the policy class induced by what each module can observe, remember, and communicate. Section 4 extends this logic to learning by using a quotient of environments. The statistical quotient describes what feedback can reveal. The joint quotient describes what regret still depends on.

Alignment feedback and reward models. RLHF and related preference-learning pipelines train a feedback model, then optimize a policy against that model [7, 26, 30, 36]. Constitutional AI and rule-based or principle-based feedback make the mediation layer more explicit [4]. Recent work shows that reward models can be brittle, miscalibrated, and vulnerable to overoptimization or spurious features such as response length [10, 11, 20, 22, 27, 31]. MG is orthogonal to reward-model accuracy. Even with a fixed true objective, a mediated interface can erase or misroute distinctions that determine the optimal response.

Partial observability, information complexity, and learning. Bayesian and information-directed views of reinforcement learning connect regret to posterior uncertainty and information gain [17, 21, 32]. Recent work studies representation learning for POMDPs and the role of information structure in partially observable teams and games [3, 38]. Our dynamic MG formalizes a narrower object. The learner does not need to identify the full environment. It needs the quotient induced by the fixed mediation architecture.

Routing and modular systems. Sparse modular computation appears in mixture-of-experts models, routing architectures, and modern reasoning pipelines [6, 8, 9, 15, 19, 25, 34, 40]. MG gives a decision-theoretic lens for such systems. A router or guardrail is valuable when it routes the distinctions that move the optimal action. More modules, more bits, or more intermediate traces can fail to reduce MG if they do not reach the component that makes the final decision.

C Implications for Modern Systems

Guardrail mediated generation. A guardrail pipeline is a direct instance of mediation. The base model proposes or drafts content. A safety classifier, verifier, or policy filter evaluates risk. An orchestrator decides which signal is visible, whether clarification is requested, and how safety evidence affects the final response. Structural MG measures the value lost when this interface hides

or blocks action-changing evidence. Dynamic MG measures how much preserved capacity remains unused after finite training.

Reward modeling. MG suggests that reward models should prioritize decision-relevant distinctions. A feature can explain large variance in human scores and still be low value if it rarely changes the best action. Length bias is one example. It can strongly affect reward-model scores while offering weak guidance near the safety boundary [20, 22, 35]. The relevant question is not whether a signal is predictive in general. The relevant question is whether it separates actions that the deployed system would otherwise confuse.

Constitutional and verifier pipelines. Principles, critiques, and verifiers decompose alignment into intermediate judgments [4, 15, 25]. MG separates two failure modes in such pipelines. A verifier may fail structurally because it lacks the decisive evidence. It may also fail dynamically because the decisive evidence exists but the learner has not discovered how to use it. This distinction matters for debugging. Adding a stronger verifier can reduce structural MG while increasing dynamic MG if the resulting interface becomes harder to learn.

Mixture-of-experts and modular routing. MoE systems route tokens or hidden states to a sparse subset of experts [8, 9, 34, 40]. From the MG view, routing quality is not only about load balance or throughput. It is about whether the routed representation carries the distinctions that the selected expert needs. A routing channel that increases capacity but sends evidence to the wrong module can preserve structural value in principle and still create a large dynamic MG at finite training budget.

Evaluation and system design. The framework recommends reporting both structural and dynamic quantities. A system can look worse after training while having a better structural interface, or look better because its weaker interface is easier to saturate. This matters for alignment evaluation. Comparing realized scores alone can penalize architectures that preserve more capacity but need more interaction to exploit it. MG therefore suggests a two-axis evaluation protocol. Measure what the interface permits in principle. Then measure how much of that capacity the learner realizes.

D Proofs for Section 2

D.1 Formal setup

We work on a filtered probability space $(\Omega, \mathcal{F}, \mathbb{P})$. Let H_t denote the full history up to time t . For each architecture $\mathcal{A}$, each agent m , and each time t , let

$$\mathcal{I}_t^{m,\mathcal{A}} \subseteq \sigma(H_t)$$

denote the information available to agent m at time t .

The policy class induced by $\mathcal{A}$ is

$$\Pi_{\mathcal{A}} = \left\{ \pi = (\pi_t^m)_{m,t} \mid \pi_t^m \text{ is } \mathcal{I}_t^{m,\mathcal{A}}\text{-measurable for every } m, t \right\}.$$

Fix a reference policy class Π^{ref} and a target objective J . Define

$$J^{\text{ref},\star} = \sup_{\pi \in \Pi^{\text{ref}}} J(\pi), \quad J_{\mathcal{A}}^{\star} = \sup_{\pi \in \Pi_{\mathcal{A}}} J(\pi).$$

For a learning trajectory $\pi_1, \dots, \pi_T \in \Pi_{\mathcal{A}}$, define

$$\bar{J}_T(\mathcal{A}) = \frac{1}{T} \sum_{t=1}^T J(\pi_t).$$

Then

$$\text{MG}_T(\mathcal{A} \mid \Pi^{\text{ref}}) = J^{\text{ref},\star} - \bar{J}_T(\mathcal{A}) = (J^{\text{ref},\star} - J_{\mathcal{A}}^{\star}) + (J_{\mathcal{A}}^{\star} - \bar{J}_T(\mathcal{A})).$$

D.2 General nesting principle

Proposition 3 (Filtration nesting implies policy class nesting). *If*

$$\mathcal{I}_t^{m,\mathcal{A}_2} \subseteq \mathcal{I}_t^{m,\mathcal{A}_1} \quad \text{for all } m, t,$$

then

$$\Pi_{\mathcal{A}_2} \subseteq \Pi_{\mathcal{A}_1}.$$

596 *Proof.* Take any policy $\pi \in \Pi_{\mathcal{A}_2}$. By definition of $\Pi_{\mathcal{A}_2}$, for every agent m and every time t , the local
 597 decision rule π_t^m is $\mathcal{I}_t^{m, \mathcal{A}_2}$ -measurable.

598 Fix any pair (m, t) . Since

$$\mathcal{I}_t^{m, \mathcal{A}_2} \subseteq \mathcal{I}_t^{m, \mathcal{A}_1},$$

599 the σ -algebra $\mathcal{I}_t^{m, \mathcal{A}_2}$ is a sub- σ -algebra of $\mathcal{I}_t^{m, \mathcal{A}_1}$. A standard fact from measure theory states that if
 600 a map is measurable with respect to a smaller σ -algebra, then it is also measurable with respect to
 601 any larger σ -algebra containing it. Hence π_t^m is also $\mathcal{I}_t^{m, \mathcal{A}_1}$ -measurable.

602 Since this argument holds for every m and every t , it follows that $\pi \in \Pi_{\mathcal{A}_1}$. Therefore

$$\Pi_{\mathcal{A}_2} \subseteq \Pi_{\mathcal{A}_1}.$$

603

□

604 D.3 Observation nesting

605 **Proposition 4** (Observation nesting). *Let two observation structures O_1 and O_2 induce local obser-*
 606 *vation histories*

$$Y_{0:t}^{m, (1)}, \quad Y_{0:t}^{m, (2)}.$$

607 Assume that for every agent m and time t , there exists a measurable map g_t^m such that

$$Y_{0:t}^{m, (2)} = g_t^m(Y_{0:t}^{m, (1)}) \quad \text{almost surely.}$$

608 Assume also that the remaining components of the local information sets are identical across the
 609 two architectures. More precisely, assume that local action histories, admissible randomness, and
 610 communication histories are the same under both architectures. Define

$$\mathcal{I}_t^{m, (i)} = \sigma(Y_{0:t}^{m, (i)}, A_{0:t-1}^m, U^m, M_{0:t}^m),$$

611 where U^m denotes the allowed randomization source and $M_{0:t}^m$ denotes the full message history
 612 available to agent m up to time t , which is assumed identical under the two observation structures.

613 Then

$$\mathcal{I}_t^{m, (2)} \subseteq \mathcal{I}_t^{m, (1)} \quad \text{for all } m, t,$$

614 and therefore

$$\Pi_{O_2} \subseteq \Pi_{O_1}.$$

615 *Proof.* Fix an agent m and a time t . By assumption, there exists a measurable map g_t^m such that

$$Y_{0:t}^{m, (2)} = g_t^m(Y_{0:t}^{m, (1)}) \quad \text{a.s.}$$

616 This implies that the random variable $Y_{0:t}^{m, (2)}$ is measurable with respect to the σ -algebra

$$\sigma(Y_{0:t}^{m, (1)}).$$

617 Equivalently,

$$\sigma(Y_{0:t}^{m, (2)}) \subseteq \sigma(Y_{0:t}^{m, (1)}).$$

618 Now enlarge both sides by the same auxiliary variables that are assumed identical under the two
 619 observation structures. Since adjoining common random variables preserves inclusion of generated
 620 σ -algebras, we obtain

$$\sigma(Y_{0:t}^{m, (2)}, A_{0:t-1}^m, U^m, M_{0:t}^m) \subseteq \sigma(Y_{0:t}^{m, (1)}, A_{0:t-1}^m, U^m, M_{0:t}^m).$$

621 That is,

$$\mathcal{I}_t^{m, (2)} \subseteq \mathcal{I}_t^{m, (1)}.$$

622 Since the argument holds for every m and t , Proposition 3 applies and yields

$$\Pi_{O_2} \subseteq \Pi_{O_1}.$$

623

□

Remark 1 (Stochastic garbling). A more general statement can be given when $Y^{(2)}$ is obtained from $Y^{(1)}$ by a conditional stochastic kernel. Suppose that, conditional on $Y_{0:t}^{m,(1)}$, the variable $Y_{0:t}^{m,(2)}$ is sampled from a kernel $K_t^m(\cdot \mid Y_{0:t}^{m,(1)})$. If the richer architecture allows enough internal randomization to simulate this kernel, then any policy admissible under O_2 can be implemented under O_1 by first simulating $Y^{(2)}$ from $Y^{(1)}$ and then applying the O_2 -policy. In that case the inclusion

$$\Pi_{O_2} \subseteq \Pi_{O_1}$$

still holds. We omit the full formalization because the measurable-factor case above is sufficient for the main text.

631 **D.4 Coordination nesting**

Proposition 5 (Coordination nesting). Let $G_1 = (V, E_1)$ and $G_2 = (V, E_2)$ be two directed communication graphs with

$$E_2 \subseteq E_1.$$

634 Assume the following.

- 635 1. The local observation structures are identical under the two graphs.
- 636 2. For every common edge $e \in E_2$, the message alphabet, delivery timing, and protocol semantics are identical under G_1 and G_2 .
- 637 3. Every extra edge $e \in E_1 \setminus E_2$ admits a distinguished null message $m_\emptyset$.
- 638 4. Policies under the richer graph G_1 may ignore extra incoming messages and may send $m_\emptyset$ on extra outgoing edges.

641 Let $\mathcal{I}_t^{m,G_i}$ denote the filtration generated by local observations, local action histories, admissible randomness, and all messages received by agent m up to time t under graph G_i . Then

$$\mathcal{I}_t^{m,G_2} \subseteq \mathcal{I}_t^{m,G_1} \quad \text{for all } m, t,$$

643 and therefore

$$\Pi_{G_2} \subseteq \Pi_{G_1}.$$

644 *Proof.* Fix an agent m and a time t . Under graph G_2 , the messages available to agent m up to time t are those received over the incoming edges of m that belong to E_2 . Under graph G_1 , the available messages include those same messages because $E_2 \subseteq E_1$.

647 Let $M_{0:t}^{m,G_i}$ denote the full message history received by agent m up to time t under graph G_i . By assumptions (1) and (2), the local observations and the common-edge message protocols are identical under the two graphs. Assumption (3) provides a null message on every extra edge, and assumption (4) says that policies under G_1 may both ignore these extra incoming messages and use the null message on extra outgoing edges. Hence any information available under G_2 is reproducible under G_1 , while the extra information under G_1 can be discarded.

653 Formally, the σ -algebra generated by the common observations, action histories, randomization sources, and G_2 -message history is contained in the larger σ -algebra generated by the same common variables together with the full G_1 -message history:

$$\sigma(Y_{0:t}^m, A_{0:t-1}^m, U^m, M_{0:t}^{m,G_2}) \subseteq \sigma(Y_{0:t}^m, A_{0:t-1}^m, U^m, M_{0:t}^{m,G_1}).$$

656 Thus

$$\mathcal{I}_t^{m,G_2} \subseteq \mathcal{I}_t^{m,G_1}.$$

657 Since this holds for every m and t , Proposition 3 implies

$$\Pi_{G_2} \subseteq \Pi_{G_1}.$$

658 □

659 D.5 Structural ladders

660 **Corollary 3** (Structural ladders). *Suppose*

$$\mathcal{A}^{(0)} \preceq \mathcal{A}^{(1)} \preceq \dots \preceq \mathcal{A}^{(L)},$$

661 *which means*

$$\mathcal{I}_t^{m, \mathcal{A}^{(\ell-1)}} \subseteq \mathcal{I}_t^{m, \mathcal{A}^{(\ell)}} \quad \text{for all } m, t, \ell = 1, \dots, L.$$

662 *Then*

$$\Pi_{\mathcal{A}^{(0)}} \subseteq \Pi_{\mathcal{A}^{(1)}} \subseteq \dots \subseteq \Pi_{\mathcal{A}^{(L)}},$$

663 *and therefore*

$$J_{\mathcal{A}^{(0)}}^* \leq J_{\mathcal{A}^{(1)}}^* \leq \dots \leq J_{\mathcal{A}^{(L)}}^*.$$

664 *Moreover, for every $k < L$,*

$$J_{\mathcal{A}^{(L)}}^* - J_{\mathcal{A}^{(k)}}^* = \sum_{\ell=k+1}^L \left(J_{\mathcal{A}^{(\ell)}}^* - J_{\mathcal{A}^{(\ell-1)}}^* \right).$$

665 *Proof.* The inclusion

$$\Pi_{\mathcal{A}^{(\ell-1)}} \subseteq \Pi_{\mathcal{A}^{(\ell)}}$$

666 for each ℓ follows from Proposition 3. Chaining these inclusions yields the full nesting relation

$$\Pi_{\mathcal{A}^{(0)}} \subseteq \Pi_{\mathcal{A}^{(1)}} \subseteq \dots \subseteq \Pi_{\mathcal{A}^{(L)}}.$$

667 Now let $\ell \in \{1, \dots, L\}$. Since

$$\Pi_{\mathcal{A}^{(\ell-1)}} \subseteq \Pi_{\mathcal{A}^{(\ell)}},$$

668 taking suprema of the same objective J over nested sets gives

$$J_{\mathcal{A}^{(\ell-1)}}^* = \sup_{\pi \in \Pi_{\mathcal{A}^{(\ell-1)}}} J(\pi) \leq \sup_{\pi \in \Pi_{\mathcal{A}^{(\ell)}}} J(\pi) = J_{\mathcal{A}^{(\ell)}}^*.$$

669 Thus the class optima are monotone along the ladder.

670 Finally, the telescoping identity is purely algebraic:

$$\sum_{\ell=k+1}^L \left(J_{\mathcal{A}^{(\ell)}}^* - J_{\mathcal{A}^{(\ell-1)}}^* \right) = (J_{\mathcal{A}^{(k+1)}}^* - J_{\mathcal{A}^{(k)}}^*) + (J_{\mathcal{A}^{(k+2)}}^* - J_{\mathcal{A}^{(k+1)}}^*) + \dots + (J_{\mathcal{A}^{(L)}}^* - J_{\mathcal{A}^{(L-1)}}^*),$$

671 and all intermediate terms cancel, leaving

$$J_{\mathcal{A}^{(L)}}^* - J_{\mathcal{A}^{(k)}}^*.$$

672

□

673 E Appendix for Section 3

674 E.1 Standing assumptions

675 The following regularity conditions is assumed.

676 **Assumption A1.** The history space $\mathcal{H}$, action space $\mathcal{A}$, and all signal spaces are standard Borel spaces.

678 **Assumption A2.** The reward function $r : \mathcal{H} \times \mathcal{A} \rightarrow [-R_{\max}, R_{\max}]$ is measurable and bounded. The discount factor satisfies $\gamma \in (0, 1)$.

680 **Assumption A3.** The reference class is the centralized causal class Π_{cent} , consisting of joint policies with full history access. The corresponding Bellman optimal value functions V^*, Q^* exist, are measurable, and satisfy

$$V^*(h) = \sup_{a \in \mathcal{A}} Q^*(h, a),$$

683

$$Q^*(h, a) = r(h, a) + \gamma \int V^*(h') P(dh' \mid h, a),$$

684 where $P(\cdot \mid h, a)$ is the history transition kernel induced by the environment.

685 **Assumption A4.** The initial history distribution is μ_0 . The return of a policy π is

$$J(\pi) = \mathbb{E}_{H_0 \sim \mu_0, \pi} \left[\sum_{t=0}^{\infty} \gamma^t r(H_t, A_t) \right].$$

686 The reference value is

$$J^* = \sup_{\pi \in \Pi_{\text{cent}}} J(\pi) = \mathbb{E}_{H_0 \sim \mu_0} [V^*(H_0)].$$

687 **Assumption A5.** In Proposition 1, the correspondence $z \mapsto \mathcal{Q}_\phi(z) \subseteq \Delta(\mathcal{A})$ is nonempty and
688 measurable, so that the conditional Bayes value is a measurable function of z .

689 **Remark.** Assumptions A1–A4 are standard in reinforcement learning and infinite horizon dis-
690 counted control. They ensure that all expectations are finite and that the Bellman objects used in the
691 theorem statements are well defined.

692 E.2 Proof of Proposition 1

693 Recall

$$J_\phi^* = \mathbb{E} \left[\sup_{q \in \mathcal{Q}_\phi(Z)} \sum_a q(a) \mathbb{E}[U(a, H) \mid Z] \right].$$

694 For each realized mediated signal $Z = z$, define the conditional Bayes value

$$B_\phi(z) := \sup_{q \in \mathcal{Q}_\phi(z)} \sum_a q(a) \mathbb{E}[U(a, H) \mid Z = z].$$

695 By Assumption A5, $B_\phi(z)$ is measurable. Then

$$J_\phi^* = \mathbb{E}[B_\phi(Z)].$$

696 Since

$$\delta_\phi(z) = \mathbb{E}[V^*(H) \mid Z = z] - B_\phi(z),$$

697 the law of total expectation gives

$$\mathbb{E}[\delta_\phi(Z)] = \mathbb{E}[V^*(H)] - \mathbb{E}[B_\phi(Z)] = \mathbb{E}[V^*(H)] - J_\phi^*.$$

698 This proves Proposition 1. □

699 **Remark.** If $\mathcal{Q}_\phi(z) = \Delta(\mathcal{A})$ for every z , then

$$B_\phi(z) = \max_a \mathbb{E}[U(a, H) \mid Z = z],$$

700 so the toy regret becomes the usual conditional Bayes regret for the compressed signal Z . This is the
701 classical value of information form associated with comparison of experiments and dynamic decision
702 making [5, 17].

703 E.3 Proof of Theorem 1

704 For any delegated policy $\pi \in \Pi_a$, define

$$d^*(H_t, A_t) = V^*(H_t) - Q^*(H_t, A_t).$$

705 By Assumption A3,

$$Q^*(H_t, A_t) = r(H_t, A_t) + \gamma \mathbb{E}[V^*(H_{t+1}) \mid H_t, A_t].$$

706 Hence

$$d^*(H_t, A_t) = V^*(H_t) - r(H_t, A_t) - \gamma \mathbb{E}[V^*(H_{t+1}) \mid H_t, A_t].$$

707 Taking expectation under π ,

$$\mathbb{E}_\pi[d^*(H_t, A_t)] = \mathbb{E}_\pi[V^*(H_t)] - \mathbb{E}_\pi[r(H_t, A_t)] - \gamma \mathbb{E}_\pi[V^*(H_{t+1})].$$

708 Multiply by γ^t and sum from $t = 0$ to T :

$$\sum_{t=0}^T \gamma^t \mathbb{E}_\pi[d^*(H_t, A_t)] = \sum_{t=0}^T \gamma^t \mathbb{E}_\pi[V^*(H_t)] - \sum_{t=0}^T \gamma^t \mathbb{E}_\pi[r(H_t, A_t)] - \sum_{t=0}^T \gamma^{t+1} \mathbb{E}_\pi[V^*(H_{t+1})].$$

709 The two value sums telescope:

$$\sum_{t=0}^T \gamma^t \mathbb{E}_\pi[V^*(H_t)] - \sum_{t=0}^T \gamma^{t+1} \mathbb{E}_\pi[V^*(H_{t+1})] = \mathbb{E}_\pi[V^*(H_0)] - \gamma^{T+1} \mathbb{E}_\pi[V^*(H_{T+1})].$$

710 Therefore

$$\sum_{t=0}^T \gamma^t \mathbb{E}_\pi[d^*(H_t, A_t)] = \mathbb{E}_\pi[V^*(H_0)] - \sum_{t=0}^T \gamma^t \mathbb{E}_\pi[r(H_t, A_t)] - \gamma^{T+1} \mathbb{E}_\pi[V^*(H_{T+1})].$$

711 By bounded rewards and $\gamma \in (0, 1)$, V^* is bounded by $R_{\max}/(1 - \gamma)$, so the last term vanishes as
712 $T \rightarrow \infty$. Using Assumption A4,

$$\mathbb{E}_\pi[V^*(H_0)] = J^*,$$

713 and

$$\sum_{t=0}^{\infty} \gamma^t \mathbb{E}_\pi[r(H_t, A_t)] = J(\pi).$$

714 Hence

$$J^* - J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

715 This proves Theorem 1. □

716 **Remark.** Theorem 1 is a history space performance difference identity. The same telescoping
717 argument underlies classical policy improvement and trust region analysis in reinforcement learning
718 [21, 32].

719 E.4 Proof of Corollary 1

720 For every $\pi \in \Pi_a$, Theorem 1 gives

$$J^* - J(\pi) = \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

721 Taking the infimum over $\pi \in \Pi_a$,

$$\inf_{\pi \in \Pi_a} (J^* - J(\pi)) = \inf_{\pi \in \Pi_a} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

722 Since J^* is constant in π ,

$$\inf_{\pi \in \Pi_a} (J^* - J(\pi)) = J^* - \sup_{\pi \in \Pi_a} J(\pi) = J^* - J_a^*.$$

723 This proves Corollary 1. □

724 E.5 Proof of Theorem 2

725 By definition,

$$\text{MG}_{\text{str}}(R) = J^* - \sup_{a \in \mathcal{A}(R)} J_a^*.$$

726 For any family of real numbers $\{x_a\}_{a \in \mathcal{A}(R)}$,

$$c - \sup_a x_a = \inf_a (c - x_a).$$

727 Applying this identity with $c = J^*$,

$$\text{MG}_{\text{str}}(R) = \inf_{a \in \mathcal{A}(R)} (J^* - J_a^*).$$

728 Now use Corollary 1 for each architecture a :

$$J^* - J_a^* = \inf_{\pi \in \Pi_a} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

729 Substituting gives

$$\text{MG}_{\text{str}}(R) = \inf_{a \in \mathcal{A}(R)} \inf_{\pi \in \Pi_a} \mathbb{E}_\pi \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

730 This proves Theorem 2. □

Remark. The term endogenous refers to the feedback loop created by the architecture. The architecture determines the admissible policy class. The chosen policy generates the trajectory. The regret distortion is evaluated on that induced trajectory.

E.6 Proof of Corollary 2

The equivalence of statements (1) and (2) follows directly from Theorem 2. For the stronger existence statement, assume the infimum is attained by some $a^* \in \mathcal{A}(R)$ and $\pi^* \in \Pi_{a^*}$, and that

$$\text{MG}_{\text{str}}(R) = 0.$$

Then

$$0 = \mathbb{E}_{\pi^*} \left[\sum_{t=0}^{\infty} \gamma^t d^*(H_t, A_t) \right].$$

Each summand is nonnegative. Therefore

$$d^*(H_t, A_t) = 0 \quad \text{almost surely for every } t.$$

Conversely, if such a^*, π^* exist, then the discounted regret distortion is zero, hence Theorem 2 gives $\text{MG}_{\text{str}}(R) = 0$.

If the infimum is not attained, statement (2) still has the right operational meaning. It says there exists a sequence of feasible architecture policy pairs whose discounted oracle regret converges to zero. $\square$

E.7 Static specialization

The one step toy problem in Section 3.1 is recovered by setting the horizon to one decision and identifying

$$r(H, A) = U(A, H), \quad \gamma = 0.$$

Then

$$V^*(H) = \max_a U(a, H), \quad Q^*(H, a) = U(a, H),$$

so

$$d^*(H, a) = V^*(H) - U(a, H).$$

Theorem 1 becomes

$$\mathbb{E}[V^*(H)] - \mathbb{E}[U(A, H)] = \mathbb{E}[d^*(H, A)],$$

and Proposition 1 is exactly the conditional Bayes decomposition of this one step regret under the mediated signal Z .

F Appendix for Section 4

F.1 Formal setup and standing assumptions

Fix an architecture a and its feasible policy class Π_a . We model repeated deployment as an episodic Bayesian decision problem. Each round deploys one policy $\pi \in \Pi_a$, produces one observable feedback variable Y , and is evaluated by its true discounted value $J_\theta(\pi)$.

Let $(\Theta, \mathcal{T})$ be the environment space, $(\Pi_a, \mathcal{P}_a)$ the measurable policy space, and $(\mathcal{Y}, \mathcal{G})$ the measurable feedback space. Let μ_0 be a prior on $(\Theta, \mathcal{T})$. For every $\theta \in \Theta$ and $\pi \in \Pi_a$, let K_θ^π be a probability measure on $(\mathcal{Y}, \mathcal{G})$. We view K_θ^π as the law of the entire feedback summary returned after one deployment episode under policy π in environment θ . Let $J_\theta(\pi) \in \mathbb{R}$ be the corresponding episode value.

A learner L is a sequence of stochastic kernels

$$L_n(d\pi \mid f_{n-1}), \quad n \geq 1,$$

from past transcripts

$$f_{n-1} = (\pi_1, y_1, \dots, \pi_{n-1}, y_{n-1}) \in (\Pi_a \times \mathcal{Y})^{n-1}$$

to $(\Pi_a, \mathcal{P}_a)$. Given $\Theta = \theta$ and learner L , the law of the transcript

$$F_T = (\pi_1, Y_1, \dots, \pi_T, Y_T)$$

is generated recursively by

$$\pi_n \sim L_n(\cdot \mid F_{n-1}), \quad Y_n \sim K_\theta^{\pi_n}.$$

By the Ionescu–Tulcea theorem, this defines a unique joint law $P_\theta^{L,T}$ on $(\Pi_a \times \mathcal{Y})^T$.

We use the following standing assumptions.

767 **Assumption B1.** The measurable spaces $(\Theta, \mathcal{T})$, $(\Pi_a, \mathcal{P}_a)$, and $(\mathcal{Y}, \mathcal{G})$ are standard Borel.

768 **Assumption B2.** The map $(\theta, \pi) \mapsto J_\theta(\pi)$ is jointly measurable and bounded. There exists
 769 $J_{\max} < \infty$ such that

$$|J_\theta(\pi)| \leq J_{\max} \quad \text{for all } (\theta, \pi) \in \Theta \times \Pi_a.$$

770 **Assumption B3.** The map $(\theta, \pi) \mapsto K_\theta^\pi(B)$ is jointly measurable for every $B \in \mathcal{G}$. Equivalently,
 771 $(\theta, \pi) \mapsto K_\theta^\pi$ is a measurable stochastic kernel from $\Theta \times \Pi_a$ to $\mathcal{Y}$.

772 **Assumption B4.** Round n feedback depends on the environment and the current deployed policy
 773 only. Formally, for every $n \geq 1$ and every $B \in \mathcal{G}$,

$$P(Y_n \in B \mid \Theta, \pi_{1:n}, Y_{1:n-1}) = K_\Theta^{\pi_n}(B) \quad \text{a.s.}$$

774 **Assumption B5.** The statistical quotient Z_a^{stat} and the joint quotient Z_a , both equipped with their
 775 quotient σ -fields, are standard Borel. The quotient maps

$$T_a^{\text{stat}}: \Theta \rightarrow Z_a^{\text{stat}}, \quad T_a: \Theta \rightarrow Z_a$$

776 are measurable.

777 **Assumption B6.** The pushforward measures $T_{a\#}^{\text{stat}} \mu_0$ and $T_{a\#} \mu_0$ admit regular conditional distri-
 778 butions along the quotient maps. In particular, there exists a measurable stochastic kernel

$$\Xi_0(dz' \mid z)$$

779 from Z_a^{stat} to Z_a such that $\Xi_0(\cdot \mid z)$ is supported on the fiber $g_a^{-1}(z)$, where g_a is the refinement map
 780 introduced below.

781 **Assumption B7.** The joint quotient value envelope

$$J_a^*(z) = \sup_{\pi \in \Pi_a} J_z(\pi)$$

782 is measurable and bounded on Z_a .

783 Assumptions B1–B7 are standard for episodic Bayesian decision problems on measurable spaces.
 784 They ensure that transcript laws, quotient posteriors, information quantities, and Bayes envelopes are
 785 well defined.

786 F.2 Auxiliary lemmas

787 **Lemma 1** (Refinement map). *The joint quotient refines the statistical quotient. Equivalently, there
 788 exists a measurable map*

$$g_a: Z_a \rightarrow Z_a^{\text{stat}}$$

789 such that

$$T_a^{\text{stat}} = g_a \circ T_a.$$

790 *Proof.* If $\theta \sim_a \theta'$, then by definition $\theta \approx_a \theta'$. Hence every $\sim_a$ -equivalence class is contained in
 791 one $\approx_a$ -equivalence class. This induces a well defined map from joint classes to statistical classes.
 792 Measurability follows from the quotient construction in Assumption B5. $\square$

793 **Lemma 2** (Adaptive transcript law). *Fix $T \geq 1$ and a learner L . Under Assumptions B1–B4, the law
 794 of the transcript under environment θ is*

$$P_\theta^{\mathsf{L}, T}(df_{1:T}) = \prod_{n=1}^T L_n(d\pi_n \mid f_{1:n-1}) K_\theta^{\pi_n}(dy_n).$$

795 Consequently, if

$$K_\theta^\pi = K_{\theta'}^\pi \quad \text{for every } \pi \in \Pi_a,$$

796 then

$$P_\theta^{\mathsf{L}, T} = P_{\theta'}^{\mathsf{L}, T} \quad \text{for every learner } \mathsf{L} \text{ and every } T \geq 1.$$

797 *Proof.* The first identity is the recursive construction of the transcript law under Assumption B4. The
 798 second claim follows immediately because the stage wise kernels coincide at every round, hence the
 799 product construction yields the same joint law. $\square$

800 **F.3 Proof of Proposition 2**

801 *Proof.* Let $\Theta \in \{\theta^+, \theta^-\}$ with prior μ_0 , and assume

$$K_{\theta^+}^\pi = K_{\theta^-}^\pi \quad \text{for every } \pi \in \Pi_a.$$

802 By Lemma 2, for every learner L and every n , the law of F_{n-1} is the same under θ^+ and θ^- . Hence
803 the conditional law of the deployed policy π_n given F_{n-1} is also the same in both environments.

804 Fix n . Using the tower property and the definition of δ_a^{toy} ,

$$\begin{aligned} & \mathbb{E}_{\Theta \sim \mu_0, L} [J_{a, \Theta}^* - J_\Theta(\pi_n)] \\ &= \mathbb{E} \left[\sum_{s \in \{+, -\}} \mu_0(\theta^s) (J_{a, \theta^s}^* - J_{\theta^s}(\pi_n)) \right] \\ &\geq \inf_{\pi \in \Pi_a} \sum_{s \in \{+, -\}} \mu_0(\theta^s) (J_{a, \theta^s}^* - J_{\theta^s}(\pi)) \\ &= \delta_a^{\text{toy}}. \end{aligned}$$

805 Averaging over $n = 1, \dots, T$ yields

$$\mathcal{D}_{a, T}(L; \mu_0) \geq \delta_a^{\text{toy}}.$$

806 For the positivity claim, suppose no single policy $\pi \in \Pi_a$ attains J_{a, θ^+}^* and J_{a, θ^-}^* simultaneously.
807 Then for every $\pi \in \Pi_a$,

$$\sum_{s \in \{+, -\}} \mu_0(\theta^s) (J_{a, \theta^s}^* - J_{\theta^s}(\pi)) > 0.$$

808 Taking the infimum over π gives $\delta_a^{\text{toy}} > 0$. □

809 **F.4 Proof of Theorem 3**

810 *Proof.* Let

$$S = T_a^{\text{stat}}(\Theta) \in Z_a^{\text{stat}}.$$

811 If two environments θ, θ' satisfy $T_a^{\text{stat}}(\theta) = T_a^{\text{stat}}(\theta')$, then $\theta \approx_a \theta'$. By Lemma 2,

$$P_\theta^{L, T} = P_{\theta'}^{L, T} \quad \text{for every learner } L \text{ and every } T.$$

812 Therefore the transcript law depends on θ only through S . Equivalently,

$$P(F_T \in \cdot \mid \Theta = \theta) = P(F_T \in \cdot \mid S = T_a^{\text{stat}}(\theta)).$$

813 This gives the conditional independence

$$F_T \perp\!\!\!\perp \Theta \mid S.$$

814 Hence

$$I(\Theta; F_T \mid S) = 0,$$

815 and the chain rule yields

$$I(\Theta; F_T) = I(S; F_T).$$

816 Now let

$$Z = T_a(\Theta) \in Z_a.$$

817 By Lemma 1, $S = g_a(Z)$. Since the transcript law already depends on Θ only through S , it follows
818 that

$$F_T \perp\!\!\!\perp Z \mid S.$$

819 Therefore

$$I(Z; F_T) = I(S; F_T) + I(Z; F_T \mid S) = I(S; F_T).$$

820 Combining the two identities gives

$$I(\Theta; F_T) = I(Z_a^{\text{stat}}; F_T) \geq I(Z_a; F_T).$$

821 □

822 **F.5 Proof of Theorem 4**

823 *Proof.* Let $Z = T_a(\Theta)$. Because $\sim_a$ requires both statistical equivalence and value equivalence, the
 824 quantities

$$K_z^\pi \quad \text{and} \quad J_z(\pi)$$

825 are well defined for every $z \in Z_a$ and every $\pi \in \Pi_a$. Let $\bar{\mu}_0 = T_{a\#}\mu_0$ be the pushforward prior on
 826 Z_a .

827 Under the quotient problem, the learner L interacts with latent environment $Z \sim \bar{\mu}_0$, observes
 828 feedback generated from K_Z^π , and is scored by $J_Z(\pi)$. Since the original and quotient models induce
 829 the same transcript law and the same values on each deployment,

$$J_{a,\Theta}^* = J_a^*(T_a(\Theta)), \quad J_\Theta(\pi_n) = J_{T_a(\Theta)}(\pi_n) \quad \text{a.s.}$$

830 Therefore

$$\begin{aligned} \mathcal{D}_{a,T}(L; \mu_0, \Theta) &= \frac{1}{T} \mathbb{E} \left[\sum_{n=1}^T (J_{a,\Theta}^* - J_\Theta(\pi_n)) \right] \\ &= \frac{1}{T} \mathbb{E} \left[\sum_{n=1}^T (J_a^*(Z) - J_Z(\pi_n)) \right] \\ &= \mathcal{D}_{a,T}(L; \bar{\mu}_0, Z_a). \end{aligned}$$

831

□

832 **F.6 Proof of Theorem 5**

833 *Proof.* Let

$$Z = T_a(\Theta) \quad \text{and} \quad \nu_{n-1} = \mathbb{P}(Z \in \cdot \mid F_{n-1}).$$

834 Fix n . Since the learner chooses π_n from F_{n-1} , we have

$$\pi_n \perp\!\!\!\perp Z \mid F_{n-1}.$$

835 Hence

$$\begin{aligned} &\mathbb{E}[J_a^*(Z) - J_Z(\pi_n) \mid F_{n-1}, \pi_n] \\ &= \int (J_a^*(z) - J_z(\pi_n)) \nu_{n-1}(dz) \\ &\geq \inf_{\pi \in \Pi_a} \int (J_a^*(z) - J_z(\pi)) \nu_{n-1}(dz) \\ &= \delta_a(\nu_{n-1}). \end{aligned}$$

836 Taking expectations and summing over n gives

$$\begin{aligned} \mathcal{D}_{a,T}(L; \mu_0) &= \frac{1}{T} \sum_{n=1}^T \mathbb{E}[J_a^*(Z) - J_Z(\pi_n)] \\ &\geq \frac{1}{T} \sum_{n=1}^T \mathbb{E}[\delta_a(\nu_{n-1})]. \end{aligned}$$

837

□

838 **F.7 Proof of Theorem 6**

839 *Proof.* Let

$$S = T_a^{\text{stat}}(\Theta) \in Z_a^{\text{stat}} \quad \text{and} \quad Z = T_a(\Theta) \in Z_a.$$

840 By Theorem 3, the transcript law depends on Θ only through S . Since Z is a function of Θ and
 841 $S = g_a(Z)$, we obtain

$$F_{n-1} \perp\!\!\!\perp Z \mid S \quad \text{for every } n \geq 1.$$

842 Therefore Assumption B6 implies

$$\mathbb{P}(Z \in \cdot \mid S, F_{n-1}) = \mathbb{P}(Z \in \cdot \mid S) = \Xi_0(\cdot \mid S) \quad \text{a.s.}$$

843 Fix n . Conditioning on S, F_{n-1} , and π_n , we have

$$\begin{aligned} & \mathbb{E}[J_a^*(Z) - J_Z(\pi_n) \mid S, F_{n-1}, \pi_n] \\ &= \int_{g_a^{-1}(S)} (J_a^*(z') - J_{z'}(\pi_n)) \Xi_0(dz' \mid S) \\ &\geq \inf_{\pi \in \Pi_a} \int_{g_a^{-1}(S)} (J_a^*(z') - J_{z'}(\pi)) \Xi_0(dz' \mid S) \\ &= b_a(S). \end{aligned}$$

844 Taking expectations and averaging over n yields

$$\mathcal{D}_{a,T}(\mathbb{L}; \mu_0) \geq \mathbb{E}[b_a(S)] = \mathbb{E}[b_a(Z_a^{\text{stat}})].$$

845

□

846 F.8 A collapse criterion

847 **Proposition 6** (Visible value collapse). *Assume that for every policy $\pi \in \Pi_a$ there exists a bounded*
 848 *measurable function $\rho_\pi: \mathcal{Y} \rightarrow \mathbb{R}$ such that*

$$J_\theta(\pi) = \int_{\mathcal{Y}} \rho_\pi(y) K_\theta^\pi(dy) \quad \text{for every } \theta \in \Theta.$$

849 Then

$$\approx_a = \sim_a.$$

850 Consequently,

$$Z_a^{\text{stat}} = Z_a$$

851 up to measurable relabeling, and the hidden value floor in Theorem 6 vanishes.

852 *Proof.* If $\theta \approx_a \theta'$, then $K_\theta^\pi = K_{\theta'}^\pi$ for every $\pi \in \Pi_a$. Hence

$$J_\theta(\pi) = \int \rho_\pi(y) K_\theta^\pi(dy) = \int \rho_\pi(y) K_{\theta'}^\pi(dy) = J_{\theta'}(\pi) \quad \text{for every } \pi \in \Pi_a.$$

853 Thus $\theta \sim_a \theta'$. The reverse implication is immediate from the definition of $\sim_a$. Therefore $\approx_a = \sim_a$,
 854 which implies equality of the quotient spaces up to measurable relabeling. If the two quotients
 855 coincide, every fiber $g_a^{-1}(z)$ is a singleton, so $b_a(z) = 0$ for every z . □

856 F.9 An achievability route

857 For the attainable frontier from Section 4.5, a converse alone does not determine the rate. One needs
 858 an explicit learner whose round wise regret is controlled by the information it extracts from feedback.
 859 The next theorem records one generic route.

860 **Theorem 7** (Achievability under a quotient information ratio). *Let*

$$S = Z_a^{\text{stat}} \quad \text{and} \quad Z = Z_a.$$

861 Fix a learner $\mathbb{L}^*$. For each round n , define

$$\nu_{n-1} = \mathbb{P}(Z \in \cdot \mid F_{n-1}),$$

862 and

$$\Delta_a(\nu_{n-1}, \pi_n) = \int (J_a^*(z) - J_z(\pi_n)) \nu_{n-1}(dz).$$

863 Assume there exists $\Gamma_a < \infty$ such that, for every $n \geq 1$,

$$\Delta_a(\nu_{n-1}, \pi_n)^2 \leq \Gamma_a I(S; Y_n \mid F_{n-1}, \pi_n) \quad \text{a.s.}$$

864 Then

$$\mathcal{D}_{a,T}(\mathbb{L}^*; \mu_0) \leq \sqrt{\frac{\Gamma_a I(S; F_T)}{T}}.$$

865 In particular, if $I(S; F_T) \leq B$, then

$$\mathcal{D}_{a,T}(\mathbb{L}^*; \mu_0) \leq \sqrt{\frac{\Gamma_a B}{T}}.$$

866 *Proof.* By the same conditioning argument used in the proof of Theorem 5,

$$\mathcal{D}_{a,T}(\mathbf{L}^*; \mu_0) = \frac{1}{T} \sum_{n=1}^T \mathbb{E}[\Delta_a(\nu_{n-1}, \pi_n)].$$

867 Apply Cauchy–Schwarz,

$$\begin{aligned} \mathcal{D}_{a,T}(\mathbf{L}^*; \mu_0) &\leq \frac{1}{T} \sqrt{T \sum_{n=1}^T \mathbb{E}[\Delta_a(\nu_{n-1}, \pi_n)^2]} \\ &\leq \frac{1}{T} \sqrt{T \Gamma_a \sum_{n=1}^T \mathbb{E}[I(S; Y_n \mid F_{n-1}, \pi_n)]}. \end{aligned}$$

868 Since the learner chooses π_n using F_{n-1} only,

$$I(S; \pi_n \mid F_{n-1}) = 0.$$

869 Hence the chain rule for mutual information gives

$$I(S; F_T) = \sum_{n=1}^T I(S; Y_n \mid F_{n-1}, \pi_n).$$

870 Substituting into the previous display yields

$$\mathcal{D}_{a,T}(\mathbf{L}^*; \mu_0) \leq \sqrt{\frac{\Gamma_a I(S; F_T)}{T}}.$$

871 If $I(S; F_T) \leq B$, the stated bound follows immediately. $\square$

872 **G Appendix for Section 5**

873 **G.1 Setup and assumptions**

874 We work in cost convention. Higher value J corresponds to lower cost $L = -J$. Let $(\Omega, \mathcal{F}, \mathbb{P})$ carry
875 a full local episode state X and a mediated view $Z = \phi(X)$. Let $h(X) \in \mathbb{R}^d$ be square integrable. A
876 controller is a square integrable measurable map into $\mathbb{R}^d$.

877 **Assumption 1** (Quadratic local surrogate). *There is a local surrogate loss*

$$\tilde{\ell}_X(u) = \frac{1}{2} u^\top H u + h(X)^\top u,$$

878 where

$$H = H^\top \succ 0, \quad \mathbb{E} \|h(X)\|_2^2 < \infty.$$

879 *Sparse coordination is encoded by a full column rank matrix $P_E \in \mathbb{R}^{d \times k}$. The sparse control space*
880 *is*

$$\mathcal{U}_E = \text{Im}(P_E).$$

881 Since $H \succ 0$ and P_E has full column rank,

$$P_E^\top H P_E \succ 0.$$

882 Define the expected surrogate cost

$$\tilde{L}(u) = \mathbb{E}[\tilde{\ell}_X(u(X))].$$

883 The three surrogate controller classes are

$$\mathcal{U}_{\text{dense}} = L^2(\sigma(X); \mathbb{R}^d),$$

884

$$\mathcal{U}_{\text{sparse}} = L^2(\sigma(X); \text{Im}(P_E)),$$

885

$$\mathcal{U}_{\text{partial}} = L^2(\sigma(Z); \text{Im}(P_E)).$$

886 Dense means full local control. Sparse means restricted coordination directions. Partial means the
887 same sparse control must be measurable with respect to the mediated view.

888 Let

$$K_E = P_E(P_E^\top H P_E)^{-1} P_E^\top.$$

889 The operator K_E is symmetric positive semidefinite. It is not generally the Euclidean projection onto
890 $\text{Im}(P_E)$. The identity used below is

$$K_E H K_E = K_E.$$

891 **Assumption 2** (Uniform local surrogate accuracy). *Let L be the true expected local cost. Let*

$$\delta(u) = L(u) - \tilde{L}(u).$$

892 *For a set $\mathcal{N}$ containing the controllers used below,*

$$|\delta(u)| \leq \eta \quad \text{for all } u \in \mathcal{N}.$$

893 The main text uses Assumption 2. We also record a smoother perturbation condition.

894 **Assumption 3** (Smooth local surrogate accuracy). *On a local Hilbert neighborhood $\mathcal{N}$, the true cost
895 L is Fréchet differentiable, μ -strongly convex, and β -smooth. Moreover,*

$$\|\nabla L(u) - \nabla \tilde{L}(u)\|_{L^2} \leq \varepsilon \quad \text{for all } u \in \mathcal{N}.$$

896 G.2 Closed form surrogate optimizers

897 **Lemma 3** (Sparse quadratic minimizer). *For fixed $h \in \mathbb{R}^d$, the minimizer of*

$$u \mapsto \frac{1}{2} u^\top H u + h^\top u$$

898 *over $u \in \mathbb{R}^d$ is $-H^{-1}h$. The minimizer over $u \in \text{Im}(P_E)$ is $-K_E h$.*

899 *Proof.* The full minimizer satisfies

$$H u + h = 0,$$

900 hence $u = -H^{-1}h$.

901 For the sparse problem, write $u = P_E v$. The restricted loss is

$$\frac{1}{2} v^\top (P_E^\top H P_E) v + (P_E^\top h)^\top v.$$

902 The first order condition is

$$(P_E^\top H P_E) v + P_E^\top h = 0.$$

903 Thus

$$v = -(P_E^\top H P_E)^{-1} P_E^\top h,$$

904 and

$$u = P_E v = -P_E (P_E^\top H P_E)^{-1} P_E^\top h = -K_E h.$$

905

□

906 Let

$$m(Z) = \mathbb{E}[h(X) \mid Z].$$

907 **Lemma 4** (Dense, sparse, and partial optima). *Under Assumption 1, the surrogate optima over the
908 three classes are*

$$\begin{aligned} u_d(X) &= -H^{-1}h(X), \\ u_s(X) &= -K_E h(X), \\ u_p(Z) &= -K_E m(Z). \end{aligned}$$

911 *Proof.* The dense and sparse formulas follow pointwise from Lemma 3. For the partial class, fix Z .

912 The conditional surrogate risk for a Z -measurable sparse action $u \in \text{Im}(P_E)$ is

$$\mathbb{E}[\tilde{\ell}_X(u) \mid Z] = \frac{1}{2} u^\top H u + \mathbb{E}[h(X) \mid Z]^\top u = \frac{1}{2} u^\top H u + m(Z)^\top u.$$

913 Applying Lemma 3 with $h = m(Z)$ yields

$$u_p(Z) = -K_E m(Z).$$

914

□

915 **G.3 Structural terms**

916 **Proposition 7** (Coordination term). *Define*

$$C = \mathbb{E}[\tilde{\ell}_X(u_s(X)) - \tilde{\ell}_X(u_d(X))].$$

917 *Then*

$$C = \frac{1}{2} \mathbb{E}[h(X)^\top (H^{-1} - K_E) h(X)].$$

918 *In particular,*

$$C \leq \frac{1}{2} \|H^{-1} - K_E\|_{\text{op}} \mathbb{E}\|h(X)\|_2^2.$$

919 *Proof.* For fixed X , $u_d(X)$ is the full minimizer. Hence

$$\tilde{\ell}_X(u_s) - \tilde{\ell}_X(u_d) = \frac{1}{2} (u_s - u_d)^\top H (u_s - u_d).$$

920 Equivalently, using minimized values,

$$\tilde{\ell}_X(u_d) = -\frac{1}{2} h(X)^\top H^{-1} h(X),$$

921 and

$$\tilde{\ell}_X(u_s) = -\frac{1}{2} h(X)^\top K_E h(X).$$

922 Subtracting gives

$$\tilde{\ell}_X(u_s) - \tilde{\ell}_X(u_d) = \frac{1}{2} h(X)^\top (H^{-1} - K_E) h(X).$$

923 Taking expectations proves the identity. The operator norm bound follows from

$$h^\top (H^{-1} - K_E) h \leq \|H^{-1} - K_E\|_{\text{op}} \|h\|_2^2.$$

924

□

925 **Proposition 8** (Information term). *Define*

$$I = \mathbb{E}[\tilde{\ell}_X(u_p(Z)) - \tilde{\ell}_X(u_s(X))].$$

926 *Then*

$$I = \frac{1}{2} \mathbb{E}[(h(X) - m(Z))^\top K_E (h(X) - m(Z))].$$

927 *Equivalently,*

$$I = \frac{1}{2} \text{tr}(K_E \mathbb{E}[\text{Cov}(h(X) \mid Z)]).$$

928 *Proof.* Both $u_s(X)$ and $u_p(Z)$ lie in $\text{Im}(P_E)$. Since $u_s(X)$ is the sparse full information minimizer,

$$\tilde{\ell}_X(u_p) - \tilde{\ell}_X(u_s) = \frac{1}{2} (u_p - u_s)^\top H (u_p - u_s).$$

929 By Lemma 4,

$$u_p - u_s = -K_E m(Z) + K_E h(X) = K_E (h(X) - m(Z)).$$

930 Using $K_E H K_E = K_E$,

$$\tilde{\ell}_X(u_p) - \tilde{\ell}_X(u_s) = \frac{1}{2} (h(X) - m(Z))^\top K_E (h(X) - m(Z)).$$

931 Taking expectations proves the first identity.

932 For the covariance form, condition on Z . Since $m(Z) = \mathbb{E}[h(X) \mid Z]$,

$$\mathbb{E}[(h - m)^\top K_E (h - m) \mid Z] = \text{tr}(K_E \text{Cov}(h(X) \mid Z)).$$

933 Taking expectations over Z gives the result.

□

934 **Corollary 4** (Surrogate structural MG). *Under Assumption 1,*

$$\tilde{L}(u_p) - \tilde{L}(u_d) = C + I.$$

935 *Proof.* By definition,

$$\tilde{L}(u_p) - \tilde{L}(u_d) = [\tilde{L}(u_s) - \tilde{L}(u_d)] + [\tilde{L}(u_p) - \tilde{L}(u_s)].$$

936 The two terms are C and I .

□

937 **G.4 Dynamic surrogate residual**

938 Let a learner produce a square integrable Z -measurable estimate $\hat{m}_T(Z)$. Define

$$\hat{u}_T(Z) = -K_E \hat{m}_T(Z).$$

939 **Proposition 9** (Quadratic dynamic residual). *Conditional on the learner output $\hat{m}_T$,*

$$D_T^{\text{quad}} := \tilde{L}(\hat{u}_T) - \tilde{L}(u_p) = \frac{1}{2} \mathbb{E}[(\hat{m}_T(Z) - m(Z))^\top K_E (\hat{m}_T(Z) - m(Z))].$$

940 *Proof.* Fix Z . The conditional sparse partial risk is

$$q_Z(u) = \frac{1}{2} u^\top H u + m(Z)^\top u, \quad u \in \text{Im}(P_E).$$

941 Its minimizer is $u_p(Z)$. Thus

$$q_Z(\hat{u}_T) - q_Z(u_p) = \frac{1}{2} (\hat{u}_T - u_p)^\top H (\hat{u}_T - u_p).$$

942 Since

$$\hat{u}_T - u_p = -K_E \hat{m}_T + K_E m = -K_E (\hat{m}_T - m),$$

943 we obtain

$$q_Z(\hat{u}_T) - q_Z(u_p) = \frac{1}{2} (\hat{m}_T - m)^\top K_E H K_E (\hat{m}_T - m).$$

944 Using $K_E H K_E = K_E$ and then averaging over Z proves the identity. $\square$

945 **Corollary 5** (Decision relevant quotient coordinate). *Let*

$$s_E(Z) = P_E^\top m(Z), \quad \hat{s}_{E,T}(Z) = P_E^\top \hat{m}_T(Z),$$

946 *and let*

$$H_E = P_E^\top H P_E.$$

947 *Then*

$$D_T^{\text{quad}} = \frac{1}{2} \mathbb{E}[(\hat{s}_{E,T}(Z) - s_E(Z))^\top H_E^{-1} (\hat{s}_{E,T}(Z) - s_E(Z))].$$

948 *Hence the dynamic learning target under the surrogate is $s_E(Z)$, not the full state X or the full*
 949 *vector $h(X)$.*

950 *Proof.* By definition,

$$K_E = P_E H_E^{-1} P_E^\top.$$

951 Substitute this into Proposition 9:

$$(\hat{m}_T - m)^\top K_E (\hat{m}_T - m) = (P_E^\top (\hat{m}_T - m))^\top H_E^{-1} (P_E^\top (\hat{m}_T - m)).$$

952 This is the claimed expression. $\square$

953 **Corollary 6** (Quadratic Bayes envelope). *Suppose the learning transcript F_T induces a posterior*
 954 *over the function s_E . The Bayes optimal predictor under the quadratic dynamic residual is*

$$\hat{s}_{E,T}^*(Z) = \mathbb{E}[s_E(Z) \mid F_T, Z],$$

955 *and the posterior Bayes envelope is*

$$\delta_a^{\text{quad}}(F_T) = \frac{1}{2} \mathbb{E}_Z[\text{tr}(H_E^{-1} \text{Cov}(s_E(Z) \mid F_T, Z))].$$

956 *Proof.* For each $Z = z$, Proposition 9 and Corollary 5 reduce the posterior risk to a weighted squared
 957 error in $s_E(z)$:

$$\frac{1}{2} \mathbb{E}[(\hat{s}(z) - s_E(z))^\top H_E^{-1} (\hat{s}(z) - s_E(z)) \mid F_T, Z = z].$$

958 The minimizer is the posterior mean

$$\hat{s}(z) = \mathbb{E}[s_E(z) \mid F_T, Z = z].$$

959 The minimized risk is the weighted posterior covariance trace. Averaging over Z proves the result. $\square$

960 **G.5 True cost decomposition**

961 Let

$$u_J \in \arg \min_{u \in \mathcal{U}_{\text{dense}}} L(u)$$

962 be the true dense optimum. Let

$$A = L(u_d) - L(u_J).$$

963 Define

$$\Xi_{A \otimes C} = \delta(u_s) - \delta(u_d),$$

964

$$\Xi_{A \otimes I} = \delta(u_p) - \delta(u_s),$$

965

$$\Xi_{A \otimes D} = \delta(\hat{u}_T) - \delta(u_p).$$

966 **Theorem 8** (Local true finite budget MG decomposition). *Under Assumption 1,*

$$L(\hat{u}_T) - L(u_J) = A + C + I + D_T^{\text{quad}} + \Xi_{A \otimes C} + \Xi_{A \otimes I} + \Xi_{A \otimes D}.$$

967 *Proof.* Use $L = \tilde{L} + \delta$. Insert and subtract

$$L(u_d), \quad \tilde{L}(u_d), \quad \tilde{L}(u_s), \quad \tilde{L}(u_p), \quad \tilde{L}(\hat{u}_T).$$

968 Then

$$\begin{aligned} L(\hat{u}_T) - L(u_J) &= [L(u_d) - L(u_J)] \\ &\quad + [\tilde{L}(u_s) - \tilde{L}(u_d)] \\ &\quad + [\tilde{L}(u_p) - \tilde{L}(u_s)] \\ &\quad + [\tilde{L}(\hat{u}_T) - \tilde{L}(u_p)] \\ &\quad + [\delta(u_s) - \delta(u_d)] \\ &\quad + [\delta(u_p) - \delta(u_s)] \\ &\quad + [\delta(\hat{u}_T) - \delta(u_p)]. \end{aligned}$$

969 The seven bracketed terms are, by definition,

$$A, \quad C, \quad I, \quad D_T^{\text{quad}}, \quad \Xi_{A \otimes C}, \quad \Xi_{A \otimes I}, \quad \Xi_{A \otimes D}.$$

970

□

971 **Corollary 7** (Uniform accuracy upper bound). *Assume Assumptions 1 and 2, and assume*

$$u_J, u_d, u_s, u_p, \hat{u}_T \in \mathcal{N}.$$

972 Then

$$0 \leq A \leq 2\eta,$$

973

$$|\Xi_{A \otimes C}| \leq 2\eta, \quad |\Xi_{A \otimes I}| \leq 2\eta, \quad |\Xi_{A \otimes D}| \leq 2\eta.$$

974 Consequently,

$$L(\hat{u}_T) - L(u_J) \leq C + I + D_T^{\text{quad}} + 8\eta.$$

975 For the structural case $\hat{u}_T = u_p$,

$$L(u_p) - L(u_J) \leq C + I + 6\eta.$$

976 *Proof.* Since u_J minimizes L , $A \geq 0$. Since u_d minimizes $\tilde{L}$ over the dense class,

$$\tilde{L}(u_d) - \tilde{L}(u_J) \leq 0.$$

977 Thus

$$A = L(u_d) - L(u_J) = \tilde{L}(u_d) - \tilde{L}(u_J) + \delta(u_d) - \delta(u_J) \leq \delta(u_d) - \delta(u_J) \leq 2\eta.$$

978 The three coupling bounds follow directly from

$$|\delta(u) - \delta(v)| \leq |\delta(u)| + |\delta(v)| \leq 2\eta.$$

979 Substituting these bounds into Theorem 8 gives

$$L(\hat{u}_T) - L(u_J) \leq C + I + D_T^{\text{quad}} + 8\eta.$$

980 If $\hat{u}_T = u_p$, then $D_T^{\text{quad}} = 0$ and $\Xi_{A \otimes D} = 0$, giving the structural bound.

□

981 **G.6 Smooth perturbation and Lipschitz coupling**

982 **Proposition 10** (Smooth surrogate perturbation). *Under Assumption 3, if $u_J, u_d \in \mathcal{N}$, then*

$$\|u_d - u_J\|_{L^2} \leq \frac{\varepsilon}{\mu},$$

983 *and*

$$A = L(u_d) - L(u_J) \leq \frac{\beta \varepsilon^2}{2\mu^2}.$$

984 *Proof.* The first order conditions give

$$\nabla L(u_J) = 0, \quad \nabla \tilde{L}(u_d) = 0.$$

985 Therefore

$$\|\nabla L(u_d) - \nabla L(u_J)\|_{L^2} = \|\nabla L(u_d)\|_{L^2} = \|\nabla L(u_d) - \nabla \tilde{L}(u_d)\|_{L^2} \leq \varepsilon.$$

986 Strong convexity implies strong monotonicity of the gradient:

$$\|\nabla L(u_d) - \nabla L(u_J)\|_{L^2} \geq \mu \|u_d - u_J\|_{L^2}.$$

987 Thus

$$\|u_d - u_J\|_{L^2} \leq \varepsilon / \mu.$$

988 By β -smoothness and $\nabla L(u_J) = 0$,

$$L(u_d) - L(u_J) \leq \frac{\beta}{2} \|u_d - u_J\|_{L^2}^2 \leq \frac{\beta \varepsilon^2}{2\mu^2}.$$

989

□

990 **Proposition 11** (Lipschitz coupling bound). *Assume $\delta = L - \tilde{L}$ is L_δ -Lipschitz on $\mathcal{N}$ in L^2 :*

$$|\delta(u) - \delta(v)| \leq L_\delta \|u - v\|_{L^2}.$$

991 *If $u_d, u_s, u_p, \hat{u}_T \in \mathcal{N}$, then*

$$|\Xi_{A \otimes C}| \leq L_\delta \sqrt{\frac{2C}{\lambda_{\min}(H)}},$$

992

$$|\Xi_{A \otimes I}| \leq L_\delta \sqrt{\frac{2I}{\lambda_{\min}(H)}},$$

993 *and*

$$|\Xi_{A \otimes D}| \leq L_\delta \sqrt{\frac{2D_T^{\text{quad}}}{\lambda_{\min}(H)}}.$$

994 *Consequently,*

$$\begin{aligned} & |\Xi_{A \otimes C}| + |\Xi_{A \otimes I}| + |\Xi_{A \otimes D}| \\ & \leq L_\delta \left(\sqrt{\frac{2C}{\lambda_{\min}(H)}} + \sqrt{\frac{2I}{\lambda_{\min}(H)}} + \sqrt{\frac{2D_T^{\text{quad}}}{\lambda_{\min}(H)}} \right). \end{aligned}$$

995 *Proof.* By Lipschitz continuity,

$$|\Xi_{A \otimes C}| = |\delta(u_s) - \delta(u_d)| \leq L_\delta \|u_s - u_d\|_{L^2}.$$

996 From the quadratic expansion,

$$C = \frac{1}{2} \mathbb{E}[(u_s - u_d)^\top H(u_s - u_d)] \geq \frac{\lambda_{\min}(H)}{2} \|u_s - u_d\|_{L^2}^2.$$

997 Hence

$$\|u_s - u_d\|_{L^2} \leq \sqrt{\frac{2C}{\lambda_{\min}(H)}}.$$

998 The proof for $\Xi_{A \otimes I}$ is identical using

$$I = \frac{1}{2} \mathbb{E}[(u_p - u_s)^\top H(u_p - u_s)].$$

999 The proof for $\Xi_{A \otimes D}$ is identical using

$$D_T^{\text{quad}} = \frac{1}{2} \mathbb{E}[(\hat{u}_T - u_p)^\top H(\hat{u}_T - u_p)].$$

1000 Adding the three inequalities gives the final display. □

1001 H Appendix for the Experiment

1002 H.1 Purpose and design principle

1003 The experiment is a finite, auditable implementation of the running example. We use it to test whether
 1004 a guardrail mediated generation system exhibits the same decomposition as the theory. Each episode
 1005 contains a user prompt, a hidden task mode, an initial mediated safety signal, an optional clarification
 1006 step, and a final action. The prompt pool, action set, generation model, guard model, judge model,
 1007 and scalar reward are fixed across all architectures. Only the mediation interface changes.

1008 The central empirical object is the finite budget mediation gap. For each architecture a , we separate
 1009 what the interface permits from what training realizes. The structural slice estimates the best feasible
 1010 value inside the architecture. The dynamic slice estimates the finite budget residual between that
 1011 structural best value and the value achieved by the learner. This mirrors the theoretical split between
 1012 structural MG and dynamic MG.

1013 The experiment uses Qwen 3.6 27b and Qwen 3 Guard on one H100 GPU.

1014 H.2 Episodes and reward

1015 We sample prompts from RealToxicityPrompts and stratify the sample into benign, borderline, and
 1016 toxic prompt buckets. Each prompt is converted into an episode

$$e_i = (x_i, \theta_i, g_i, c_i),$$

1017 where x_i is the prompt, $\theta_i \in \{\text{BENIGN}, \text{UNSAFE}\}$ is a hidden task mode, g_i is the initial guard
 1018 signal, and c_i is the clarification text. The hidden mode is sampled from a bucket dependent
 1019 distribution. This gives a controlled latent variable that can matter for the optimal response while
 1020 remaining hidden from weaker architectures.

1021 The final action space is

$$\mathcal{A}_{\text{final}} = \{\text{ACCEPT}, \text{REWRITE}, \text{BLOCK}\}.$$

1022 For each final action, the policy model generates a response. A guard model assigns a risk score, and
 1023 a separate judge assigns an alignment score. The scalar reward at safety weight λ is

$$R_\lambda(e_i, a) = \text{Align}(e_i, a) - \lambda \text{Risk}(e_i, a).$$

1024 The main cross-section uses $\lambda^* = 0.8$. The lambda sweep varies λ to trace how the value of safety
 1025 relevant mediation changes as unsafe responses become more costly.

1026 H.3 Architectures

1027 We compare four delegated architectures and one oracle reference. The architectures differ only in
 1028 the mediation interface.

1029 Static Binary and Static Ternary implement observation coarsening. Blocked Mediation and Routed
 1030 Mediation isolate routing. Both can ask for clarification. Only Routed Mediation allows the answer
 1031 to influence the final action. This comparison is important because it separates evidence availability
 1032 from evidence usability. A signal that exists somewhere in the system has little value if it cannot
 1033 reach the decision point.

Table 1: Architectures used in the experiment. The environment, reward, action set, and evaluation stack are fixed.

Architecture	First signal	Can clarify	Routes clarification	Latent access
Static Binary	binary severity	no	no	no
Static Ternary	ternary severity	no	no	no
Blocked Mediation	ternary severity	yes	no	no
Routed Mediation	ternary severity	yes	yes	no
Oracle	latent mode	yes	yes	yes

H.4 Policy classes

For each architecture a , we define a finite policy class Π_a . A policy table specifies a first action as a function of the architecture visible first signal. If clarification is available and requested, the table also specifies a final action as a function of the visible clarification value. Thus each Π_a is an explicit finite analogue of an architecture induced policy class.

This design lets us estimate structural optima by exhaustive search. It also lets us compare architectures without changing the underlying task. The policy class expands as the mediation interface becomes richer. Static Binary has the smallest class. Routed Mediation has a larger class because it can condition on routed clarification. The oracle class can condition directly on the latent mode.

H.5 Counterfactual outcome tensor

Before computing structural or dynamic quantities, we build a counterfactual outcome tensor. For every episode, final action, and visibility mode, we evaluate the generated response, guard score, judge score, and reward. The tensor has entries of the form

$$(e_i, v, a) \mapsto (\text{Align}(e_i, v, a), \text{Risk}(e_i, v, a), R_\lambda(e_i, v, a)),$$

where $v \in \{\text{none}, \text{clar}\}$ records whether the clarification is visible to the generation step.

The tensor is the key control device in the experiment. Once it is built, every architecture is evaluated using the same counterfactual basis. Architectural differences therefore come from the decision path and policy constraints, not from inconsistent response generation or judge calls. This gives a clean empirical analogue of comparing J_a^* across fixed architecture classes.

H.6 Structural estimator

Let $\mathcal{E}_{\text{eval}}$ be the evaluation episode set. For each architecture a , we estimate its structural best value by

$$\hat{J}_a^*(\lambda) = \max_{\pi \in \Pi_a} \frac{1}{|\mathcal{E}_{\text{eval}}|} \sum_{e_i \in \mathcal{E}_{\text{eval}}} R_\lambda(e_i, \pi).$$

The oracle reference is computed in the same discrete abstraction, using direct access to the latent mode. The structural mediation gap is

$$\widehat{\text{MG}}_{\text{str}}(a, \lambda) = \hat{J}_{\text{oracle}}^*(\lambda) - \hat{J}_a^*(\lambda).$$

This quantity measures what the architecture loses in principle, before finite training enters.

H.7 Dynamic learner and realized value

For each architecture, we train a finite budget learner over the corresponding policy class. The learner observes rewards from training rollouts and forms a posterior score for each policy. After the training budget is exhausted, it deploys the policy with the largest posterior mean on the evaluation set. We write the resulting achieved value as

$$\hat{J}_T(a, \lambda) = \frac{1}{|\mathcal{E}_{\text{eval}}|} \sum_{e_i \in \mathcal{E}_{\text{eval}}} R_\lambda(e_i, \hat{\pi}_{a,T}).$$

The realized total mediation gap is

$$\widehat{\text{MG}}_{\text{tot}}(a, \lambda) = \hat{J}_{\text{oracle}}^*(\lambda) - \hat{J}_T(a, \lambda).$$

Table 2: Cross-section of the mediation gap decomposition at $\lambda^* = 0.8$. Structural MG is the plug-in architecture optimum gap shown in Fig. 3b. Realized Total MG is estimated from seed-bootstrap intervals with $B = 5000$. Dynamic MG is Realized Total MG minus the plug-in structural base, so its interval reflects uncertainty in realized performance. The paradox indicator is positive when an architecture has larger Structural MG than Routed Mediation but lower Realized Total MG at the same finite training budget.

(a) Static architectures.

Architecture	Structural MG	Realized Total MG [95% CI]	Dynamic MG [95% CI]	Paradox vs Routed
Static Binary	0.140	0.157 [0.099, 0.225]	0.017 [-0.041, 0.085]	Yes
Static Ternary	0.140	0.303 [0.219, 0.384]	0.163 [0.079, 0.244]	Yes

(b) Dynamic architectures.

Architecture	Structural MG	Realized Total MG [95% CI]	Dynamic MG [95% CI]	Paradox vs Routed
Blocked Mediation	0.130	0.388 [0.354, 0.423]	0.258 [0.224, 0.293]	No
Routed Mediation	0.120	0.354 [0.319, 0.405]	0.234 [0.199, 0.285]	–

1064 The finite budget dynamic residual is

$$\widehat{\text{MG}}_{\text{dyn}}(a, \lambda) = \widehat{\text{MG}}_{\text{tot}}(a, \lambda) - \widehat{\text{MG}}_{\text{str}}(a, \lambda).$$

1065 This residual measures unrealized capacity inside the architecture. In the main text we use it as the
1066 empirical dynamic MG at the given training budget.

1067 H.8 Lambda sweep

1068 For each policy, the reward as a function of λ is an affine line

$$\bar{A}_\pi - \lambda \bar{R}_\pi,$$

1069 where $\bar{A}_\pi$ is mean alignment and $\bar{R}_\pi$ is mean risk. The structural value of an architecture is therefore
1070 an upper envelope of policy lines:

$$\hat{J}_a^*(\lambda) = \max_{\pi \in \Pi_a} (\bar{A}_\pi - \lambda \bar{R}_\pi).$$

1071 This representation lets us trace policy switches and structural MG as λ varies. It also explains why
1072 structural MG increases with the safety weight. When unsafe mistakes become more expensive,
1073 erased safety relevant distinctions carry a larger value penalty.

1074 H.9 Cross-section at $\lambda^* = 0.8$

1075 Table 2 gives the numerical cross-section behind the decomposition map in Fig. 3b. Structural MG is
1076 the plug-in structural base. Realized Total MG is estimated from seed-bootstrap intervals. Dynamic
1077 MG is the realized total gap minus the plug-in structural base. Its interval therefore reflects uncertainty
1078 in realized performance, while structural estimation uncertainty is held fixed.

1079 The table gives a compact view of the main phenomenon. Static Binary has a larger structural base
1080 than Routed Mediation, yet much lower realized total MG. It is structurally weaker, but easier for the
1081 learner to saturate. Routed Mediation has the smallest structural base among the delegated systems,
1082 which confirms the value of routed clarification. Its dynamic residual is large, which means the
1083 preserved capacity is difficult to realize at the present training budget.

1084 H.10 Panel interpretation

1085 Fig. 3a validates the decision landscape. The best action changes across prompt regimes, so mediation
1086 is consequential. Accepting is valuable for benign prompts and risky for toxic prompts. Rewriting
1087 is strongest in the ambiguous middle. Thus an information channel matters only when it preserves
1088 distinctions that change the optimal action.

1089 Panel A in Fig. 2 compares achieved value with the structural best at $\lambda^* = 0.8$. It shows that the
1090 finite budget learner nearly saturates Static Binary, while leaving substantial capacity unrealized in
1091 Routed and Blocked Mediation. This is the clearest visual evidence for the dynamic residual.

1092 Panel B in Fig. 2 shows the structural MG trend across λ . Structural MG rises as safety weight
1093 increases. This matches the value-weighted theory. The cost of a missing distinction depends on how
1094 much that distinction affects the current objective.

1095 Panel C in Fig. 2 traces the finite budget burden across evaluation episodes. The richer dynamic
1096 architectures carry a larger residual under the current learner. Routed Mediation improves on Blocked
1097 Mediation, which confirms that routing evidence to the decision point matters.

1098 Fig. 3b provides the decomposition map. The large number in each cell is realized total MG against
1099 the oracle. The number in parentheses is the structural base. The difference is the dynamic residual.
1100 This panel is the main empirical summary of the paper. It shows that finite-budget performance can
1101 reverse structural ordering.

1102 H.11 Interpretive insights

1103 **Weak mediation can look better after training.** Static Binary has the smallest realized total MG
1104 at $\lambda^* = 0.8$, even though it has a larger structural base than Routed Mediation. This is the finite
1105 budget paradox. A weak interface can be easier to saturate, while a richer interface can preserve
1106 capacity that the learner does not discover.

1107 **Structural richness and realized performance can diverge.** The theory predicts that total finite
1108 budget MG has both structural and dynamic components. The results show that these components can
1109 move in opposite directions. Routed Mediation lowers the structural base, while its dynamic residual
1110 remains large. This is why achieved value alone is an incomplete measure of mediation quality.

1111 **Routing is distinct from observation.** Blocked Mediation and Routed Mediation share the same
1112 clarification opportunity. They differ in whether the answer can affect the final action. Routed
1113 Mediation has a lower structural base and lower realized total MG. This shows that evidence must be
1114 usable at the decision point. Merely generating clarification is insufficient.

1115 **Clarification is useful only when it is decision-sufficient.** Routed Mediation improves over
1116 Blocked Mediation, yet its structural base remains positive. This shows that routed clarification helps,
1117 while the implemented policy class still loses some oracle-relevant distinctions. In this environment,
1118 useful mediation requires both access to the latent signal and a policy class that preserves the relevant
1119 joint state when choosing the final action.

1120 **Information value is action-switch weighted.** Panel A and the lambda sweep give the same
1121 message. Signal richness matters when it protects distinctions that change the optimal action. As
1122 the safety weight grows, unsafe mistakes become more costly, and structural MG rises. This is a
1123 value-weighted view of information, not a raw-bit view.

1124 **The dynamic residual measures unrealized capacity.** For the richer architectures, most realized
1125 total MG comes from the dynamic residual. The architecture preserves useful structure, but finite
1126 training fails to exploit it. This is the empirical analogue of dynamic MG. It motivates studying the
1127 learner and feedback channel as part of the mediation problem.

1128 **The experiment supports the decomposition.** The results do not reduce to a simple ranking of
1129 architectures. Static Binary is easiest to learn. Routed Mediation has the strongest structural capacity
1130 among delegated architectures. Blocked Mediation isolates a routing failure. The decomposition is
1131 needed to make these facts compatible. It tells us whether loss comes from the interface itself or from
1132 finite-budget learning inside the interface.

1133 I Limitation

1134 The framework has two limitations. First, the theory treats the mediation interface as known and
1135 cleanly specified. Real systems often have moving interfaces, learned routers, correlated feedback
1136 errors, and model updates that change the effective architecture during training. Second, our experi-
1137 ment is a controlled guardrail setting with finite policy tables and a stylized hidden mode. This is
1138 intentional, since it lets us isolate structural and dynamic MG without confounds, but it does not by
1139 itself establish the size of MG in frontier deployment pipelines. A direct test in open-ended multi-turn
1140 generation would require richer action spaces, online feedback, and stronger auditing of latent value
1141 distinctions.